Ecological Informatics

Integrating Multi-Sensor Earth Observation and Deep Learning for Drought Prediction and Vegetation Resilience Assessment in California

--Manuscript Draft--

Manuscript Number:	
Article Type:	Research Paper
Keywords:	Drought forecasting; Vegetation resilience; Multi-sensor remote sensing; Ecohydrology; Deep learning; Mediterranean ecosystems
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Abstract:	Drought is one of the costliest natural perils in California. We developed an integrated multi-sensor approach combining vegetation resilience mapping, ecosystem state dynamics, and deep learning forecasting to outline drought responses in California's four natural land cover classes—grasslands, natural forests, savannas, and shrublands—from 2001 to 2024. Using nine satellite-based and ground-validated drought-related indicators (NDVI, GNDVI, NIRv, GOSIF, GPP, LST, VPD, SPEI, TWS), we evaluated six deep learning models (BiLSTM, TCN, XGBoost, N-HiTS, LSTM, N-BEATS) for one-month-ahead forecasting on 216 model-index-land-type combinations. Three findings emerged. First, California ecosystems showed paradoxical resilience during 2020-2024 with substantial reductions in vegetation stress for natural forests and shrublands (8-10 percentage point improvements) relative to 2012-2016, despite terrestrial water storage remaining 2-3 standard deviations below baseline. Second, mean resilience improvements masked a bimodal split: some regions recovered while extreme stress zones expanded 2 to 17-fold, and natural forests showed 17-fold increases sufficient to shift regional carbon balances. Chronic vulnerability hotspots (0.02-0.33% of each land type) were spatially clustered, indicating reproducible failure zones requiring targeted intervention. Third, ecosystem state transitions exhibited asymmetric dynamics with stress onset probabilities 1.5-3× higher than recovery probabilities, and shrublands exhibiting 74% dry/stressed persistence, indicative of stress traps. BiLSTM achieved optimal RMSE on 94% of forecasting tasks (34 of 36), and TCN most frequently attained optimal R2 (36%). Index predictability ranged from easy (vegetation: $R2 > 0.80$) to challenging (meteorological-hydrological: $R2 < 0.40$). The findings demonstrate that recurrent drought causes ecosystem bifurcation rather than steady degradation, with threshold consequences for Mediterranean-climate ecosystems globally.

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(Formerly Faculty of Engineering)

University of Tehran

Feb, 2, 2026

Dear Editor-in-Chief,

On behalf of my co-author, I am pleased to submit our manuscript entitled "Integrating Multi-Sensor Earth Observation and Deep Learning for Drought Prediction and Vegetation Resilience Assessment in California" for consideration in the prestigious Journal of Ecological Informatics.

This study addresses a persistent gap in drought science. Operational early-warning systems often emphasize drought detection or forecasting in isolation, while ecosystem-impact studies often remain retrospective and do not quantify the dynamics that control recovery, persistence, and the emergence of chronic failure zones. We present an integrated framework that links multi-sensor drought forcing, ecosystem functional response, state-transition dynamics, and one-month-ahead forecasting across California's major natural land types over 2001–2024. We believe the following novelties make this manuscript interesting for the journal's readership:

1. A unified, multi-sensor "forcing-to-response" framework at ecosystem scale: We combine vegetation resilience mapping, ecosystem state dynamics, and deep-learning forecasting in a single workflow using nine satellite-based and ground-validated indicators. This creates a cohesive bridge between drought monitoring, impact attribution, and predictive capability- an integration that is rarely implemented systematically at landscape scale.
2. A new ecohydrologically consistent composite forcing metric (EWBI) designed for ecosystem drought assessment: We develop the Ecohydrological Water Balance Index (EWBI) to integrate meteorological water balance (SPEI-1), hydrologic storage (TWS), atmospheric demand (VPDmax), and thermal stress (LST) into one interpretable drought-forcing construct. We then use its deseasonalized form jointly with NIRv anomalies to classify ecosystem states and quantify transitions—enabling direct diagnosis of when ecosystems remain productive despite unfavorable hydroclimate conditions versus when they enter locked stress regimes.
3. A substantive new insight: recurrent drought is driving ecosystem bifurcation rather than uniform degradation: Beyond documenting mean responses, we reveal a bimodal split in ecosystem outcomes. Some regions recover while extreme-stress zones expand sharply (2–17×), with repeating spatial "failure hotspots" that are clustered and therefore actionable for targeted field investigation and management. This helps reconcile why regional averages can suggest resilience while ecologically consequential collapse zones intensify.

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4. State-transition evidence for "stress traps," hysteresis, and asymmetric recovery constraints: Using monthly transition matrices, we show that stress onset systematically exceeds recovery, and that shrublands in particular, exhibit high dry/stressed persistence (stress-trap behavior)—a mechanism-level result with clear implications for early intervention and ecosystem management under intensifying drought regimes.

5. A rigorous, large-scale benchmarking of deep-learning models for operational drought forecasting across indices and ecosystems: We evaluated six forecasting models across 216 model–index–land-type combinations under a common protocol and held-out testing, yielding clear operational guidance (e.g., BiLSTM's consistent RMSE advantages; metric-dependent strengths of alternative architectures). Importantly, we demonstrated that predictability differs fundamentally by variable class (vegetation indices vs. meteorological/hydrological indices), which is critical for designing realistic early-warning systems.

We believe the manuscript will be of strong interest to journal's readers working at the interface of remote sensing, ecohydrology, drought risk, and data-driven environmental forecasting, because it provides (i) an integrated methodology that is transferable to other regions, and (ii) decision-relevant findings that identify where drought impacts are intensifying even when mean conditions appear to recover.

This work is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission. The datasets are available from the corresponding author upon reasonable request, and the full codebase used for processing, analysis, and figure generation is publicly available via GitHub.

Sincerely,

B. Zahraie

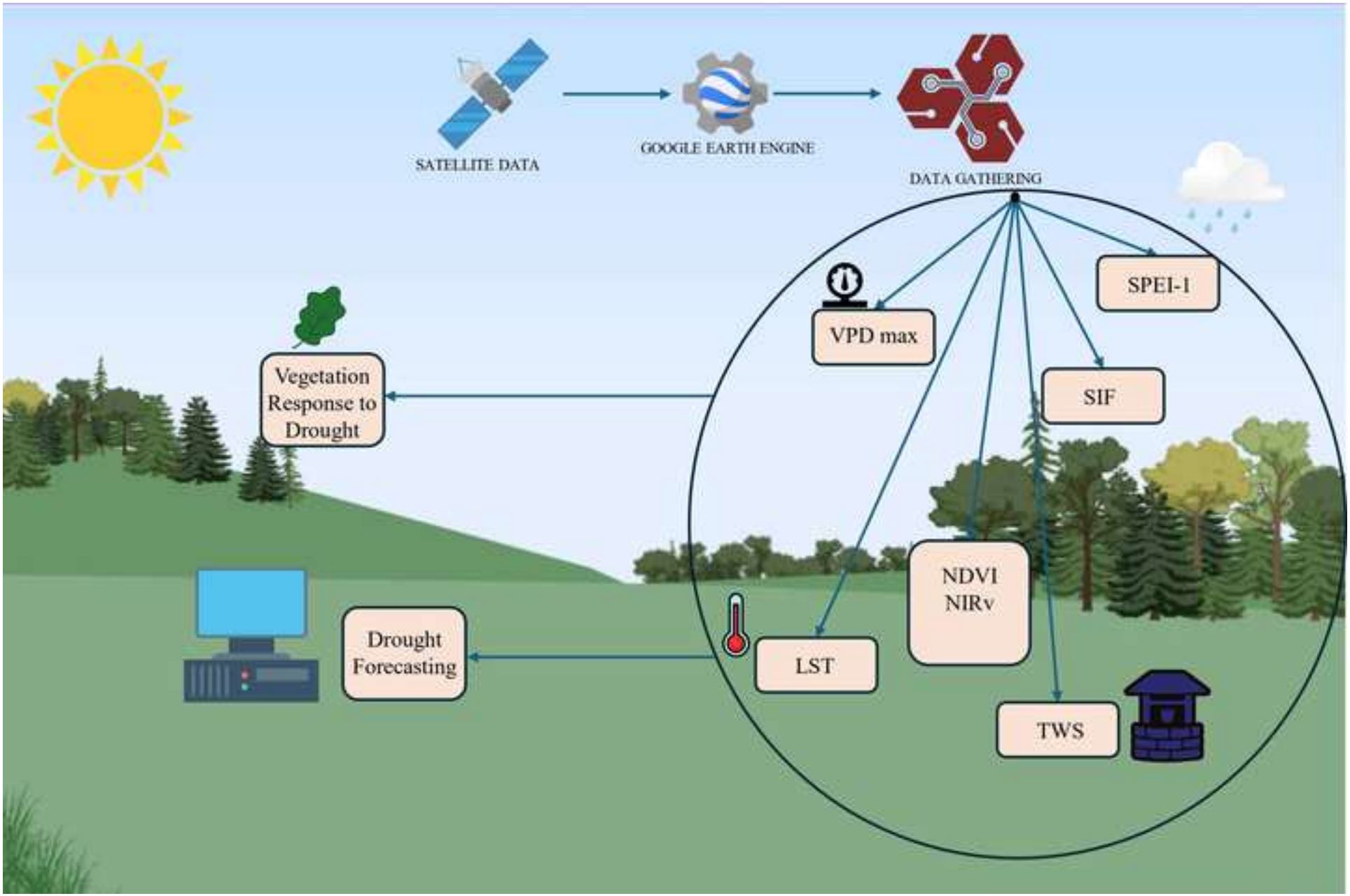
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Highlights

- Integrated multi-sensor EO to quantify drought controls on vegetation resilience.
- Developed an interpretable ML framework for monthly drought forecasting at scale.
- Provided a reproducible pipeline (GEE/Python) for monitoring drought and early warning.
- Mapped resilience hotspots and persistent stress zones at 250–500 m resolution.
- Delivers interpretable outputs suitable for management and risk communication.

Graphical Abstract

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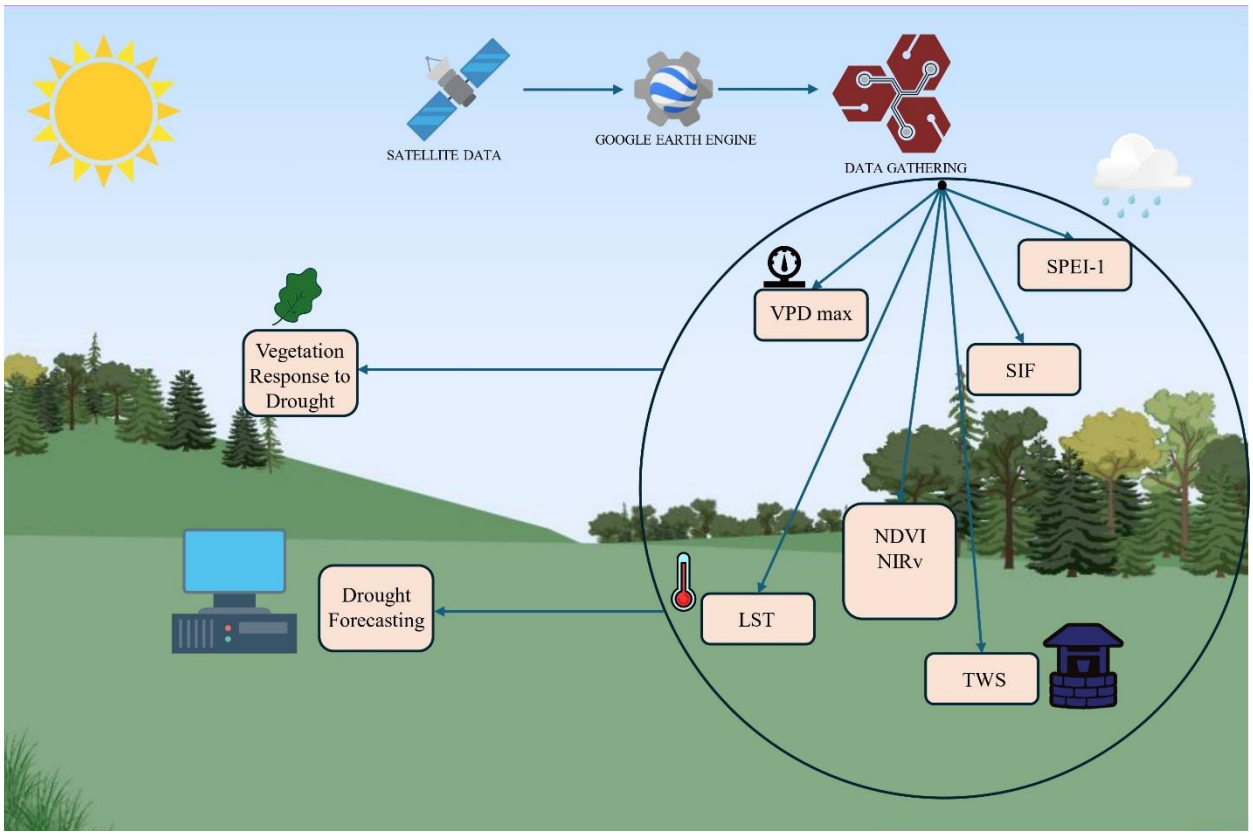
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Graphical Abstract

Abstract

Drought is one of the costliest natural perils in California. We developed an integrated multi-sensor approach combining vegetation resilience mapping, ecosystem state dynamics, and deep learning forecasting to outline drought responses in California's four natural land cover classes—grasslands, natural forests, savannas, and shrublands—from 2001 to 2024. Using nine satellite-based and ground-validated drought-related indicators (NDVI, GNDVI, NIRv, GOSIF, GPP, LST, VPD, SPEI, TWS), we evaluated six deep learning models (BiLSTM, TCN, XGBoost, N-HiTS, LSTM, N-BEATS) for one-month-ahead forecasting on 216 model-index-land-type combinations. Three findings emerged. First, California ecosystems showed paradoxical resilience during 2020-2024 with substantial reductions in vegetation stress for natural forests and shrublands (8-10 percentage point improvements) relative to 2012-2016, despite terrestrial water storage remaining 2-3 standard deviations below baseline. Second, mean resilience improvements masked a bimodal split: some regions recovered while extreme stress zones expanded 2 to 17-fold, and natural forests showed 17-fold increases sufficient to shift regional carbon balances. Chronic vulnerability hotspots (0.02-0.33% of each land type) were spatially clustered, indicating reproducible failure zones requiring targeted intervention. Third, ecosystem state transitions exhibited asymmetric dynamics with stress onset probabilities 1.5-3× higher than recovery probabilities, and shrublands exhibiting 74% dry/stressed persistence, indicative of stress traps. BiLSTM achieved optimal RMSE on 94% of forecasting tasks (34 of 36), and TCN most frequently attained optimal R^2 (36%). Index predictability ranged from easy (vegetation: $R^2 > 0.80$) to challenging (meteorological-hydrological: $R^2 < 0.40$). The findings demonstrate that recurrent drought causes ecosystem bifurcation rather than steady degradation, with threshold consequences for Mediterranean-climate ecosystems globally.

Key Words: Drought forecasting, Vegetation resilience, Multi-sensor remote sensing, Ecohydrology, Deep learning, Mediterranean ecosystems, BiLSTM

1. Introduction

Drought is one of the most financially disruptive natural disasters affecting California, particularly during recent years. The 2020–2022 drought period was the driest three consecutive years on record, causing an estimated 3.4 billion USD loss in the agricultural sector and displacing more than 34,000 workers (Medellín-Azuara et al., 2022). Major crop revenue losses reached 1.3 billion USD in 2021 and 1.7 billion USD in 2022, while downstream sectors such as food processing suffered value-added losses of 590 million USD and 845 million USD, respectively (Medellín-Azuara et al., 2022). Impacts varied markedly across regions. In 2022, surface-water deliveries in the Sacramento Valley declined by 49%, forcing more than 750,000 acres of farmland to be fallowed and increasing groundwater extraction by 27–33% relative to baseline levels (Medellín-Azuara et al., 2022). Given that California's Mediterranean-climate agriculture produces >400 commodities and generates around USD 50 billion in annual revenue and supports more than a quarter of a million jobs (Medellín-Azuara et al., 2024), drought-induced disruptions extend far beyond farm boundaries, influencing supply chains (Tchonkouang et al., 2024), groundwater

sustainability (Pauloo et al., 2020), and the viability of rural communities (Greene, 2018). These challenges underscore the need for comprehensive drought monitoring and early-warning systems that integrate ecosystem response assessment with predictive forecasting to mitigate growing risks (AghaKouchak et al., 2015; Salakpi et al., 2022).

Remote sensing plays a pivotal role in drought monitoring from regional to continental scales. (Jiao et al., 2021a). Unlike ground-based networks, satellite observations offer consistent spatial and temporal coverage. (West et al., 2019) And enable the derivation of key variables such as precipitation, land surface temperature (LST), vegetation indices, and soil moisture. (AghaKouchak et al., 2015; Jiao et al., 2021a). Multi-sensor approaches, combining optical, thermal, and microwave data, provide complementary perspectives on drought evolution, capturing the interactions among meteorological, agricultural, hydrological, and ecological processes. (Vreugdenhil et al., 2022; Zhang and Zhou, 2016). However, integrating data across sensors remains challenging due to differences in spatial and temporal resolution, units, and coverage gaps, necessitating robust data-fusion techniques for generating consistent drought assessments. (Jiao et al., 2021a).

Despite advances in monitoring, understanding ecosystem responses to drought remains incomplete. (Jiao et al., 2021a). Vegetation resilience, the capacity to maintain function during stress and recover after disturbance, is a key aspect of drought response that is still poorly quantified. (Seddon et al., 2016). Many large-scale analyses treat drought as spatially homogeneous, overlooking differences among ecosystem types. (Anderegg et al., 2016; Seddon et al., 2016). California's ecological diversity, spanning forests, grasslands, savannas, and shrublands across strong climatic and hydrological gradients (Chornesky et al., 2015; Lenihan et al., 2003; Thorne et al., 2017), provides a unique setting to assess how vegetation strategies modulate drought impacts (Malone et al., 2016). Isohydric ecosystems conserve water through stomatal closure under stress, while anisohydric systems maintain gas exchange to sustain productivity even during drought. (Novick et al., 2016). These contrasting strategies lead to distinct vulnerability and recovery behaviors that remain under-explored at landscape scales. (Chen et al., 2025; Cushman and McGarigal, 2019).

In addition to monitoring, proactive drought management requires short-term forecasting to support agricultural planning, water allocation, and wildfire risk reduction. (Hao et al., 2018; Turco et al., 2019). Machine learning (ML) and deep learning (DL) have shown strong potential for modeling nonlinear temporal dependencies in drought indices. (Dikshit et al., 2021; Park et al., 2016). Numerous drought indices exist, capturing meteorological, agricultural, hydrological, and ecological conditions. (Mishra and Singh, 2010). Yet, most studies focus on a single index or perform limited comparisons. (Hao et al., 2018)Comprehensive assessments of model performance across multiple indices and ecosystem types remain scarce. (Jiao et al., 2021b). Although advanced DL architectures such as LSTM, BiLSTM, CNN, TCN, N-HiTS, and N-BEATS have been used in geospatial forecasting (Lara-Benítez et al., 2021; Prodhan et al., 2022), systematic comparisons for drought prediction are still limited (Park et al., 2020).

Three key research gaps (RG) motivate this study:

RG1: Limited understanding of ecosystem-specific drought responses. Large-scale studies often assume spatial homogeneity and lack a detailed assessment of vegetation resilience, their ability to maintain and recover function under water stress. (Seddon et al., 2016).

RG2: Incomplete characterization of ecosystem state transitions. Shifts among healthy, stressed, suppressed, and resilient states remain poorly mapped at the landscape scale, hindering the identification of regions facing chronic ecological stress. (Kleist et al., 2025; Scheffer et al., 2001).

RG3: Lack of systematic evaluation of forecasting models and indices. Comparisons across multiple drought indices, land types, and deep-learning architectures are rare, limiting operational insights into model suitability. (Elbeltagi et al., 2024; Park et al., 2020).

Based on the above research gaps, the current study addresses the following research questions (RQ):

RQ1: How do vegetation resistance and recovery differ among California's forests, grasslands, savannas, and shrublands during the 2012–2016 drought, and which ecosystems show the highest vulnerability and adaptability?

RQ2: What are the temporal and spatial patterns of ecosystem state transitions during and after drought, and which regions exhibit chronic stress posing long-term ecological risks?

RQ3: Which deep-learning architectures yield optimal performance for one-month-ahead forecasting of different drought indices across diverse land types, and how does forecast skill vary by index and ecosystem?

1.1 Research Objectives

To address the above research questions, this study develops an integrated multi-sensor remote-sensing framework that combines vegetation resilience analysis with deep-learning-based drought forecasting. We assembled a comprehensive dataset of eight satellite-derived and one model-based ground validation drought index spanning vegetation condition (Normalized Difference Vegetation Index [NDVI], Green Normalized Difference Vegetation Index [GNDVI], Near-Infrared Reflectance of vegetation [NIRv]), ecosystem productivity (Solar-Induced Fluorescence [SIF], Gross Primary Productivity [GPP] from FLUXCOM), meteorological stress (maximum Vapor Pressure Deficit [VPDmax], 1-month Standardized Precipitation Evapotranspiration Index [SPEI-1]), thermal stress (Land Surface Temperature [LST]), and water availability (Terrestrial Water Storage [TWS]) across California's four natural land types—natural forests, grasslands, savannas, and shrublands—for the period 2001-2024 in a monthly scale. The specific objectives are as follows:

O1: To characterize land-type-specific vegetation resilience during the 2012-2016 California drought by integrating resistance and recovery indices to identify patterns of vulnerability and adaptive capacity.

O2: To quantify ecosystem state transitions among healthy, stressed, suppressed, and resilient conditions using deseasonalized anomalies, thereby mapping regions of persistent ecological stress.

O3: To systematically compare six deep-learning architectures (BiLSTM, LSTM, TCN, N-HiTS, N-BEATS, XGBoost) for one-month-ahead drought forecasting across 36 independent tasks (nine indices × four land types).

2. Study Area and Data

2.1 Study Area

This study focuses on the natural ecosystems within California state, which occupy 289,303 km² of area divided across four major land-cover categories: grasslands (109,129 km², accounting for 37.7%), natural forests (49,444 km², which is 17.1%), savannas (75,765 km², which is 26.2%), and shrublands (54,965 km², which is 19.0%) (Figure 1). The four major land-cover categories were categorized based on the 2021 MODIS Land Cover Type product (MCD12Q1 Version 6.1), which provides 500 meters of resolution (Friedl and Sulla-Menashe, 2022a). The International Geosphere-Biosphere Programme (IGBP) scheme of land-cover type has been reorganized in the following format: natural forests include evergreen needleleaf, evergreen broadleaf, deciduous needleleaf, deciduous broadleaf, and mixed forests (IGBP classes 1-5); savannas contain woody savannas and savannas (classes 8-9); grasslands include all types of grasslands (class 10); and shrublands contain both close and open shrublands (classes 6-7). The analysis did not involve the urban regions, agricultural fields, water bodies, barrens, or permanent snow and ice areas.

Table 1. Datasets and characteristics of the drought-related indicators used in this study

Data Type	Variable	Full Name	Source/Satellite	Spatial Resolution	Temporal Resolution	Time Period	Reference
Ecosystem Variables	*NIRv*	Near-Infrared Reflectance of Vegetation	MODIS (MOD13Q1 Version 6.1)	250 m	16-day	2000-present	(Didan, 2021)
	NDVI	Normalized Difference Vegetation Index	MODIS (MOD13Q1 Version 6.1)	250 m	16-day	2000-present	(Didan, 2021)
	GNDVI	Green Normalized Difference Vegetation Index	MODIS (MOD09A1.061 Terra Surface Reflectance)	500 m	8-day	2000-2025	(Vermote, 2021)
	SIF	Solar-Induced Chlorophyll Fluorescence	GOSIF v2	0.05° (~5.5 km)	8-day	2000-2024	(Li and Xiao, 2019)
	GPP	Gross Primary Productivity	FLUXCOM	0.05° (~5.5 km)	Monthly (aggregated from hourly)	2001-2021	(Nelson et al., 2023)
	GPP	Gross Primary Productivity	MODIS (MOD17A2H.061)	500 m	8-day	2021-2024	(Running et al., 2021)

Meteorological & Hydrological Variables	*SPEI*	Standardized Precipitation Evapotranspiration Index	GRIDMET DROUGHT	~4.6 km (4638.3 m)	5-day	1980-2025	(Abatzoglou, 2013)
	TWS	Terrestrial Water Storage Anomaly	GRACE and GRACE-FO (Gravity Recovery and Climate Experiment)	0.5° (~55.66 km)	Monthly	2002-2024	(Watkins et al., 2015; Wiese et al., 2016)
	VPDmax	Maximum Vapor Pressure Deficit	GRIDMET	~4 km	Daily	1979-present	(Abatzoglou, 2013)
	LST	Land Surface Temperature	MODIS/Aqua (MYD21C1 V061)	0.05° (~5.6 km)	Daily	2000-present	(Hulley and Hook, 2021)
Land Cover	*LC*	MODIS Land Cover Type	MODIS Terra and Aqua (MCD12Q1 V061)	500 m	Annual	2001-2021	(Friedl and Sulla-Menashe, 2022b)

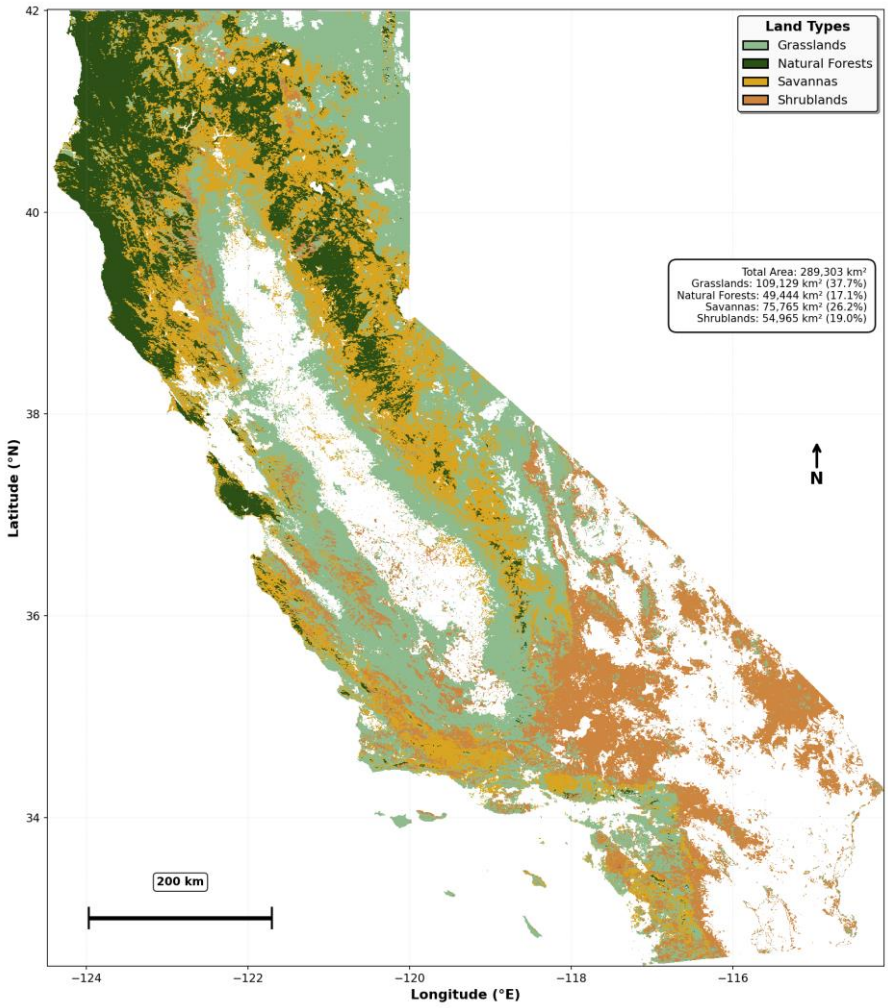


Figure 1. Spatial distribution of land types in California (2021)

2.2 Datasets, data characteristics, and sources

2.2.1 Vegetation Indices

Three vegetation indices were derived from MODIS (Moderate Resolution Imaging Spectroradiometer) satellite observations to characterize vegetation greenness and canopy condition. The Normalized Difference Vegetation Index (NDVI) and the Near-Infrared Reflectance of Vegetation (NIRv) were obtained from the MODIS Vegetation Indices product (MOD13Q1, Version 6.1) at a native spatial resolution of 250 m and a 16-day temporal composite period. (Didan, 2021). NDVI is directly provided in the MOD13Q1 product, while NIRv was computed as the product of near-infrared reflectance (ρ_{NIR}) and NDVI, following (Badgley et al., 2017), (NIRv = ρNIR × NDVI).

The Green Normalized Difference Vegetation Index (GNDVI) was calculated from the MODIS Surface Reflectance product (MOD09A1, Version 6.1) at 500 m spatial resolution and 8-day temporal composites. GNDVI was derived using the near-infrared (band 2) and green (band 4) surface reflectances, as proposed by (Gitelson et al., 1996), [GNDVI = (ρNIR - ρGreen) / (ρNIR + ρGreen)].

All vegetation indices were temporally aggregated to a monthly scale by averaging the corresponding intra-monthly composites. NDVI and GNDVI serve as general indicators of canopy greenness and photosynthetic activity, whereas NIRv provides enhanced sensitivity to canopy structure and productivity by integrating both greenness and density information.

2.2.2 Ecosystem Productivity

Two indicators of ecosystem productivity were used to represent photosynthetic activity and carbon uptake dynamics. Solar-Induced Chlorophyll Fluorescence (SIF) data were obtained from the Global OCO-2-based Solar-Induced Fluorescence (GOSIF) dataset. (Li and Xiao, 2019). This provides 8-day composites at 0.05° (~5 km) spatial resolution from 2000 to the present. GOSIF integrates OCO-2 satellite retrievals with MODIS reflectance observations to generate spatially continuous SIF estimates directly linked to the light-use efficiency of photosynthesis and vegetation physiological responses to stress.

Gross Primary Productivity (GPP) data were derived from the FLUXCOM ensemble upscaling product. (Nelson et al., 2023), which provides monthly estimates at 0.0833° (~8 km) spatial resolution for the period 2001-2021. The FLUXCOM dataset combines machine-learning models trained on FLUXNET eddy covariance tower observations with remote-sensing and meteorological inputs to estimate terrestrial carbon fluxes. Since FLUXCOM data are not available after 2021, GPP for the validation (2022–2023) and testing (2024) periods was obtained from the MODIS MOD17A2H Version 6.1 product (Running et al., 2021), an 8-day composite at 500 m resolution. MOD17A2H estimates GPP by using a light-use efficiency model driven by satellite-derived fraction of absorbed photosynthetically active radiation (FPAR), leaf area index (LAI), and meteorological variables.

Both SIF and GPP datasets were aggregated to a monthly temporal resolution and harmonized to a common 5 km spatial grid. While the two datasets differ in their methodological approach,

FLUXCOM is data-driven, and MOD17A2H is process-based (Zhao et al., 2005). They both quantify ecosystem carbon assimilation and have been shown to be spatially and temporally coherent in cross-validation studies (Joiner and Yoshida, 2020; Jung et al., 2020). Together, SIF and GPP complement greenness-based indices (NDVI, GNDVI, NIRv) by representing physiological and functional aspects of vegetation productivity rather than purely structural or spectral properties.

2.2.3 Meteorological and Hydrological Variables

Four additional indices of atmospheric tension, meteorological drought, water availability, and thermal regime were incorporated to consider the meteorological and hydrological aspects of drought. Maximum Vapor Pressure Deficit (VPDmax) was obtained from the Gridded Surface Meteorological Dataset. (Abatzoglou, 2013), which provides daily meteorological data at about 4 km spatial resolution. VPDmax represents the monthly maximum atmospheric evaporative demand and was calculated by taking the maximum daily VPD values in each month. This parameter quantifies the atmospheric deficit in moisture that triggers plant water stress by increasing transpiration demand and causing stomatal closure as a drought avoidance mechanism.

The 1-month Standardized Precipitation-Evapotranspiration Index (SPEI-1) was estimated using GRIDMET drought indices. (Abatzoglou, 2013) From the 30-day SPEI product with approximately 4 km resolution. Monthly SPEI-1 values were derived by computing the mean of daily 30-day SPEI observations within each month. SPEI-1 integrates precipitation deficiency and atmospheric evaporative demand to represent short-term meteorological drought conditions on a standardized scale, with negative values indicating drought intensity.

Terrestrial Water Storage (TWS) anomalies were derived from Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) satellite missions using the JPL Mascon RL06 solution of (Watkins et al., 2015; Wiese et al., 2016), which provides monthly estimates of equivalent liquid water thickness at an approximate 0.5° spatial resolution. Monthly TWS values were obtained by averaging all observations within each month from the lwe_thickness band. TWS represents integrated water storage in all hydrological reservoirs, like soil moisture, groundwater, surface water, and snow, with anomalies representing variations relative to the long-term baseline (2004-2009).

Land Surface Temperature (LST) was obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) Aqua sensor (MYD21C1 Version 6.1), at 0.05° spatial resolution. (Hulley and Hook, 2021). Daytime LST was sampled and converted to Celsius, with quality-controlled filtering to retain only high-quality retrievals where cloud cover <30%. LST records surface heat conditions that enhance drought effects through heightened evaporative demand and direct plant heat stress. The four indices were subsequently upscaled to a monthly time resolution and harmonized to a 5 km spatial resolution for combining with vegetation and productivity data (Figure 2).

3. Methodology

3.1 Data Preprocessing and Harmonization

After gathering all the datasets for the study, spatial harmonization was required as native resolutions between indices were not uniform. All the datasets were resampled to a common 5 km spatial resolution so that even analysis could be done. Monthly temporal aggregation was applied to all sub-monthly datasets. TWS and LST had missing values at random for a few months due to satellite data availability interruptions (GRACE to GRACE-FO transition and regular MODIS Aqua downtime) (Landerer et al., 2020; Tapley et al., 2004). We used month-wise cross-year linear interpolation to fill this: for each calendar month (all the May, for example), missing values were interpolated across years using the nearest available observations for that same month. (Moreno-Martínez et al., 2020). This preserves seasonal structure and interannual variability but restores temporal continuity to allow for additional analysis. SPEI-1 was temporally complete and did not require imputation. This approach preserved short-term interannual variability and long-term climatic trends and provided temporal continuity for subsequent analysis. SPEI-1 was not gap-filled since the dataset was temporally complete. Spatial resampling was conducted for all data in Google Earth Engine with bilinear interpolation for low-resolution data (TWS) and spatial averaging for high-resolution data (vegetation indices, LST, VPDmax) (Gorelick et al., 2017). Throughout all processing procedures, the WGS84 geographic coordinate reference system (EPSG:4326) was employed.

3.1.1 Removing the seasonality from data

We deseasonalized all nine drought indices to isolate interannual drought signals from recurring seasonal phenological cycles. Monthly anomalies were calculated by subtracting calendar-month climatological means while preserving original physical units. (Meroni et al., 2019).

For each of the drought indices, the first step was to compute the monthly climatology from the baseline period (2001-2011) by averaging the values over each calendar month of all years included in the baseline. For instance, the climatology for January was computed as the average of all January observations from 2001-2011, the February climatology from all February observations, and so on. This yielded 12 monthly climatological reference values representative of the typical seasonal conditions for each index.

Deseasonalized anomalies were then computed by subtracting the appropriate monthly climatological mean from each observation:

$$\boldsymbol{Anomaly}_{\{i,m\}} = \boldsymbol{Value}_{\{i,m\}} - \boldsymbol{Climatology}_{\boldsymbol{m}}$$

Where *i* denotes the year-month observation and *m* denotes the calendar month (1-12). This method removes seasonal cycles but retains anomalies in the original physical units of each index (e.g., dimensionless for NDVI, kPa for VPD, cm for TWS, °C for LST), enabling direct interpretation of drought magnitude in meaningful units. Positive anomalies indicate above-average conditions

relative to the climatological baseline, while negative anomalies indicate below-average conditions.

This method of deseasonalization was applied consistently to all land types and indices to create the standardized anomaly time series displayed in Figures 2 and 3. The baseline period of 2001-2011 was selected to represent pre-drought reference conditions, preceding the major 2012-2016 California drought event.

3.1.2 Identification of drought from time series

The time series of these nine indices over the four California natural land cover types during 2001 - 2024 is shown in Figs. 2 and 3. Figure 2 describes standardized anomalies for vegetation and productivity indices (NDVI, GNDVI, NIRv, GOSIF, GPP) and clearly shows strong negative anomalies for the 2012-2016 drought period, with sustained negative deviations of -1.0 to -1.5 SD exhibited by NIRv in most land cover classes. Natural Forests consistently show the maximum seasonal amplitude, whereas Shrublands exhibit maximum interannual variability. Also, the 2020-2024 time period indicates vegetation stress signals that are substantially weaker despite continued water storage deficits. Figure 3 shows meteorological and hydrological forcing indices. In Figure 3, VPDmax has elevated atmospheric demand values of +1.5 to +2.0 SD during both drought periods. SPEI-1 shows large negative values (~ -2.0 SD) during the peak drought years (2014, 2021-2022). Most critically, TWS displays a continuous decline from 2012 through 2024 (-2.0 to -3.0 SD below baseline) without recovery, representing the most pronounced hydrological signal in the record. LST shows continuous seasonal patterns with negligible drought-specific signals.

3.1.2 Ecohydrological Water Balance Index (EWBI)

To enable integrated assessment of multi-dimensional drought conditions that incorporate meteorological, hydrological, atmospheric, and thermal stressors, we developed the Ecohydrological Water Balance Index (EWBI). EWBI integrates four interdependent drought forcing indices—SPEI-1, TWS, VPDmax, and LST—into a composite indicator that captures the synergistic effects of water scarcity (SPEI, TWS) and atmospheric/thermal stress (VPD, LST) on water balance in ecosystems.

Event-scale computation.

For each land type and period P (Baseline: 2001–2011; Historic drought: 2012–2016; Recent crisis: 2020–2024), we first computed the period mean of each component across all months in P: SPEI_P, TWS_P, VPD_P, LST_P. The EWBI for each land type and period was then calculated as:

$$EWBI_P = z(SPEI_P) + z(TWS_P) - z(VPD_P) - z(LST_P)$$

Where $z(\cdot)$ denotes spatial standardization to z-scores across pixels within the land type for that same period. SPEI and TWS are positive contributors (higher values indicating wetter conditions), and VPD and LST are negative contributors (higher values indicating greater atmospheric demand

and thermal stress). The resultant index is a dimensionless estimate of relative ecohydrological water balance within each land type.

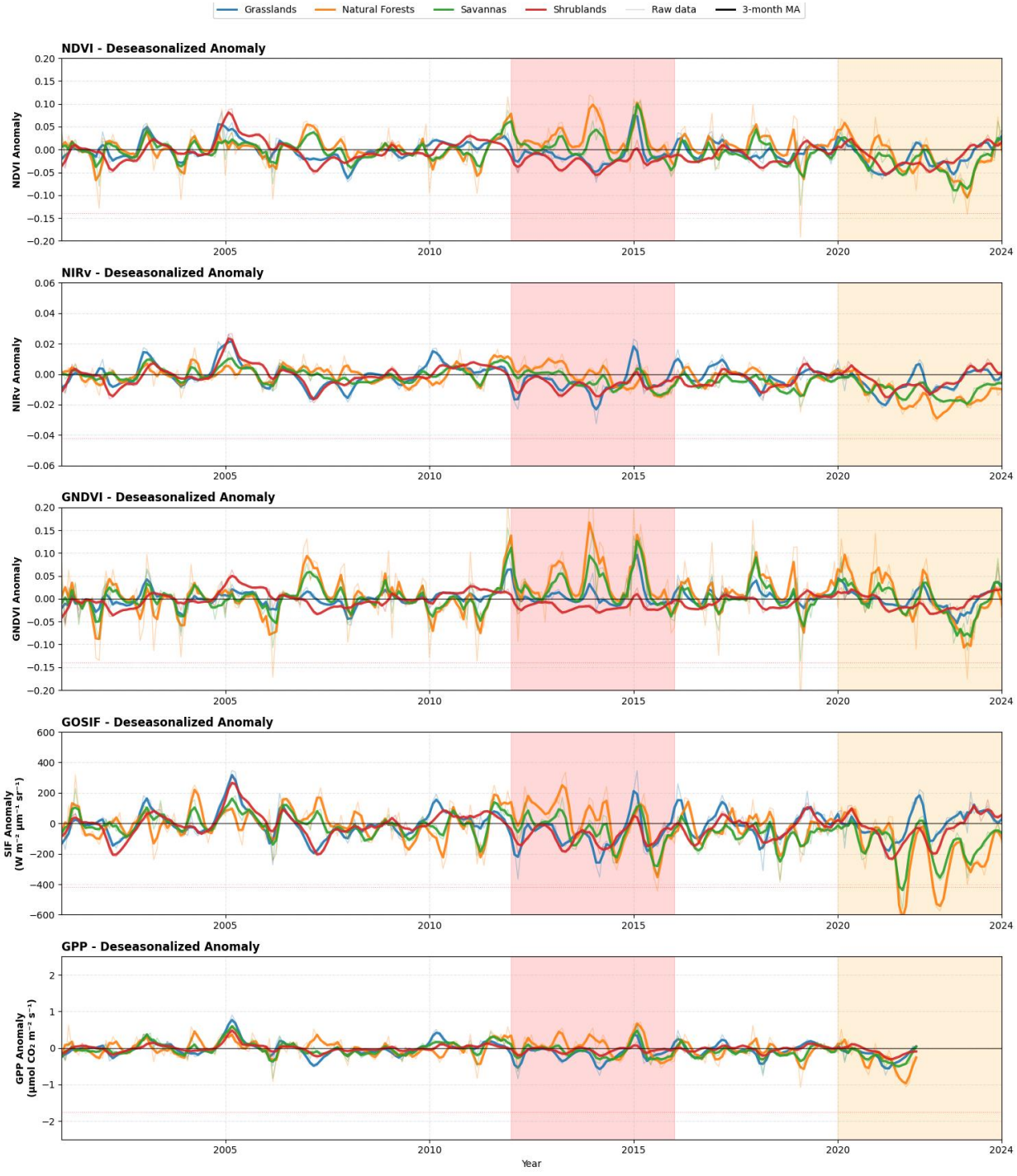


Figure 2. Vegetation and Productivity Response – Standardized Anomalies (2001-2024). Standardized anomalies (z-scores) of five vegetation and ecosystem productivity indices across

California's four natural land cover types (Natural Forests, orange; Savannas, green; Grasslands, blue; Shrublands, red). Anomalies represent deviations from monthly climatological means (2001-2011 baseline) in units of standard deviation (SD). Thin lines show raw monthly anomalies; bold lines show 3-month moving averages. Shaded areas mark the 2012-2016 drought (pink) and the 2020-2024 period (beige). Original units before standardization: NDVI, NIRv, GNDVI (dimensionless), GOSIF (W m^{-2} μm^{-1} sr^{-1}), GPP (µmol CO_2 m^{-2} s^{-1}).

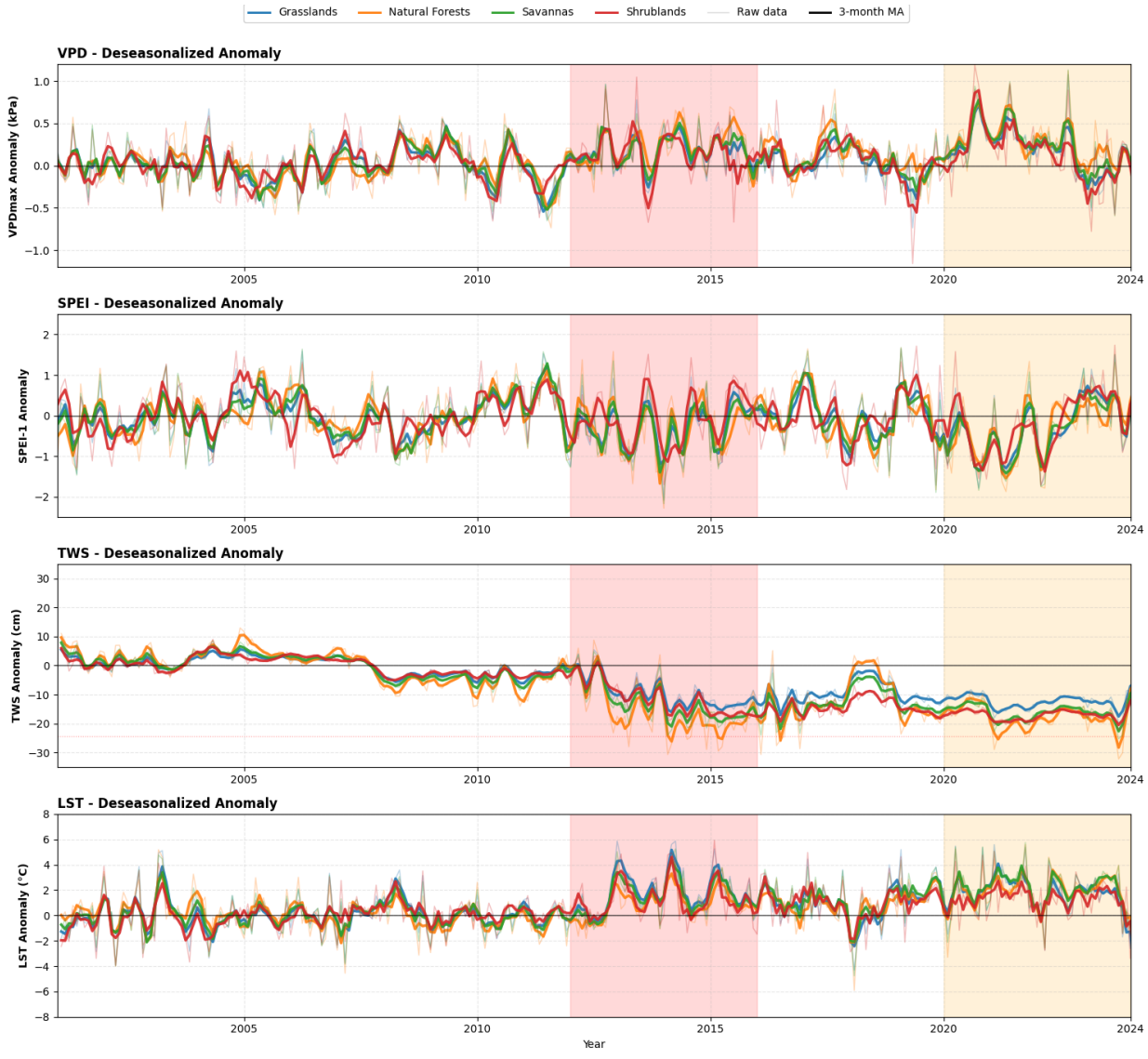


Figure 3. Meteorological and Hydrological Forcing – Standardized Anomalies (2001-2024). Standardized anomalies of four drought forcing indices follow the same format and color scheme as Figure 2. Original units before standardization: VPDmax (kPa), SPEI-1 (dimensionless), TWS (cm water equivalent), LST (°C).

Change mapping. To compare drought events, we computed differences relative to the baseline composite:

$$\Delta EWBI_{2012-2016} = EWBI_{2012-2016} - EWBI_{2001-2011},$$

$$\Delta EWBI_{2020-2024} = EWBI_{2020-2024} - EWBI_{2001-2011},$$

EWBI was only used for describing and comparing the 2012–2016 and 2020–2024 drought events and was not integrated into the prediction framework, which operates on single-index targets.

3.2 Vegetation Resilience Assessment

Vegetation resilience was evaluated with NIRv as the first-order indicator because it combines vegetation greenness and canopy structure in a single product of NDVI and near-infrared reflectance. (Badgley et al., 2017). NIRv has also higher correlations with ecosystem productivity (GPP) compared to conventional greenness indexes (NDVI, GNDVI), yielding greater sensitivity to functional drought responses instead of structural changes alone. Although NDVI is relatively invariant in response to early drought stress through compensatory processes, NIRv more clearly reflects reductions in photosynthesis and hence is optimally suited to assess ecosystem function under water limitation.

3.2.1 Definition of Baseline and Drought Period

Measuring vegetation resilience during California's mega-drought of 2012-2016, we defined a pre-drought baseline period (2001-2011) and drought period (2012-2016). Monthly NIRv anomalies were calculated by subtracting climatological means and dividing by climatological standard deviations of the baseline period, resulting in standardized z-scores to facilitate cross-ecosystem comparison (Figure 4).

3.2.2 Resistance and Recovery Metrics

Following Seddon et al. (2016), we quantified resistance (ability to maintain function under stress) and recovery (ability to return to baseline after disturbance). Resistance was defined as the minimum of the standardized NIRv anomaly for 2012-2016:

$$\text{Resistance} = \min\left(\text{NIRv}_{\text{anomaly},2012-2016}\right)$$

More negative values indicate lower resistance. Recovery was calculated as the change from the minimum drought value to the post-drought mean (2017-2019):

$$\text{Recovery} = \overline{\text{NIRv}}_{\text{anomaly},2017-2019} - \min\left(\text{NIRv}_{\text{anomaly},2012-2016}\right)$$

Positive recovery values indicate a return toward baseline. Metrics were calculated at 5 km pixel resolution for all valid pixels with ≥70% data availability in both baseline and drought periods, as determined by the Pixel Selection & Stratification module (Figure 4).

3.2.3 Ecosystem State Classification and Temporal Transitions

In order to track coupling between drought forcing and vegetation response, we specified monthly ecosystem states on the basis of the simultaneous behavior of deseasonalized EWBI (forcing composite) and deseasonalized NIRv (vegetation response). Deseasonalization was performed by subtracting calendar-month climatological medians (computed over 2001-2024) from each monthly value to isolate interannual anomalies from seasonal cycles. For every land cover, we first constructed spatial-mean monthly time series on the 5 km grid for SPEI-1, TWS, VPDmax, LST, and NIRv for the period 2001–2024. Seasonal signals were subtracted by deducting the calendar-month climatological median from each series.

From these anomalies, a monthly **EWBI′** series was constructed by time-standardizing each forcing component and combining with signs consistent with water balance:

$$EWBI'_t = z(SPEI'_t) + z(TWS'_t) - z(VPD'_t) - z(LST'_t)$$

Where z(·) denotes standardization over time (mean 0, SD 1) applied to the deseasonalized series for the given land type. States were defined using median splits of the EWBI′ and NIRv′ anomaly distributions to create binary categories (high/low) for each dimension. The median threshold for each variable was computed separately for each land type over the full 2001-2024 period:
Wet/Healthy: $EWBI' > \text{median}(EWBI')\ and\ NIRv' > median$

1. Wet/Suppressed: $EWBI' > median(EWBI')\ and\ NIRv' \leq median$
2. Dry/Resilient: $EWBI' \leq median(EWBI')\ and\ NIRv' > median$
3. Dry/Stressed: $EWBI' \leq median(EWBI')\ and\ NIRv' \leq median$

Monthly transition probabilities were calculated by counting all observed transitions from each state at month t to each state at month t+1 across the 2001-2024 period. (Marshall and Randhir, 2008). For each starting state (row), counts were converted to percentages by dividing by the total number of times that state occurred, ensuring each row sums to 100%. Diagonal values represent persistence (remaining in the same state), while off-diagonal values represent state changes.

3.3 Deep Learning Forecasting Framework

To enable operational drought prediction across California's diverse ecosystems, we set up a methodical multi-model deep learning framework for 1-month ahead univariate time series prediction. Problem formulation of forecasts, model selection strategy, architectural designs, and mathematical foundations are outlined in this section.

3.3.1 Forecasting Task Definition

The forecasting task was defined as a univariate 1-month ahead prediction for each drought index-land type combination. For a given drought index I, and land type L, we constructed a time series $X = \{x_1, x_2, ..., x_t\}$ where x_i represents the spatially aggregated mean value of index I across all pixels of land type L at month i. The forecasting objective was to predict x_{t+1} given a historical

lookback window: $x_{t+1} = f(\{x_{t-w+1}, x_{t-w+2}, \ldots, x_t\})$, where w is the lookback length (12-24 months depending on model and variable). Inputs were scaled in a model-specific manner: BiLSTM, LSTM, TCN, N-HiTS, and N-BEATS used MinMax scaling to [0,1], whereas XGBoost used z-score standardization computed from the training period's calendar-month mean and standard deviation; the fitted transforms were then applied to validation and test datasets.

For GPP, the transition from FLUXCOM (training: 2001–2021) to MODIS MOD17A2H (validation and test: 2022–2024) is a switch in methodology from flux-tower upscaling to satellite process modelling; both nevertheless quantify the same biophysical variable (gross carbon uptake) and were harmonized to the same units (gC/m²/day) through spatial and temporal preprocessing, enabling continuity in the forecasting framework. The data were temporally split into 2001–2021 for training, 2022–2023 for validation, and 2024 for testing, with all splits requiring sufficient lookback (≥12 months) to allow for model initialization. For each land category, the time series was constructed using all valid pixels (i.e., pixels with ≥30% data coverage in both the training and test intervals), and monthly land-type means were computed over this fixed set of valid pixels to ensure spatial consistency across splits. To remove long-term trends while preserving seasonal patterns, we applied month-specific detrending: for each calendar month (e.g., all Januarys, all Februarys), we fitted a separate linear trend across years (2001–2021) and subtracted it from the observed values. This approach eliminates multi-year drift (e.g., gradual warming or drying trends) while retaining the seasonal cycle that is important for forecasting. Target values were clipped to ±4.0 standard deviations to eliminate extreme outliers during model training without removing genuine drought events, which typically fall within ±3 SD. The ±4.0 SD threshold removes only statistical artifacts or data errors (<0.01% of observations) while retaining all physiologically realistic extreme conditions.

3.3.2 Model Architectures

We compared six deep learning models of three paradigms: recurrent networks (BiLSTM, LSTM), convolutional/hierarchical models (TCN, N-HiTS, N-BEATS), and tree-based ensemble methods (XGBoost). Model selection was guided by recent research providing state-of-the-art performance on time series forecasting in meteorological and ecological domains.

Bidirectional Long Short-Term Memory (BiLSTM):

We employed variable-specific configurations dependent on drought index temporal behavior and established literature-based baselines. For meteorologically-forced indices (SPEI-1), we applied a 24-month lookback with bidirectional processing with 200 LSTM units per layer to detect the multi-scale temporal relations in precipitation-evapotranspiration processes. Hydrological indices (TWS) employed an 18-month lookback with 150 bidirectional units to detect groundwater recharge cycles. Vegetation and productivity indices (NDVI, GNDVI, NIRv, GPP, GOSIF) and atmospheric stress factors (VPDmax, LST) used a 12-month lookback with 128 unidirectional LSTM units, optimized for season-to-annual variation. The model consisted of two LSTM layers with dropout regularization (0.2), followed by dense layers (10 units with ReLU activation,

followed by 1 unit with linear activation) for univariate output (Piotrowski et al., 2020). The bidirectional architecture for SPEI and TWS processes forward and backward simultaneously so that the model can leverage past and future context within the lookback window. Models were implemented in TensorFlow with GPU support and optimized using the Adam optimizer (Kingma and Ba, 2014), (learning rate = 0.001) with early stopping (patience = 10 epochs) on validation loss (Graves et al., 2005).

Long Short-Term Memory (LSTM):

We employed a straightforward unidirectional LSTM with two 128-unit hidden layers and 0.2 dropout as a baseline recurrent architecture, and a uniform 12-month lookback for all variables. This architecture provides a computational baseline against the more advanced BiLSTM and maintains the basic capability of LSTM cells to provide long-term dependence handling through gated memory. (Hochreiter and Schmidhuber, 1997).

Temporal Convolutional Network (TCN):

TCN (Bai et al., 2018), uses dilated causal convolutions to learn long-range temporal dependencies without recurrence connections, providing computational benefits with parallelized training. Our model utilized three convolutional layers with 32 channels per layer, a kernel size of 3, and exponentially expanding dilation rates (1, 2, 4) in order to increase the effective receptive field to 12 months. Residual connections and dropout (0.3) were used for gradient flow and regularization. To estimate predictive uncertainty, we applied Monte Carlo dropout during inference time (20 passes) to generate prediction intervals (Lin et al., 2020). We used mixed precision training, which combines 16-bit floating point (FP16) operations for most calculations with 32-bit floating point (FP32) operations for numerically sensitive operations. This approach reduces memory usage and accelerates training by approximately 2-3× on GPU hardware while maintaining model accuracy.

Neural Hierarchical Interpolation for Time Series (N-HiTS):

N-HiTS (Challu et al., 2023), breaks down time series into multi-rate hierarchical patterns by using stacked blocks with increasingly finer temporal pooling. Our design had three stacks with pool sizes [2, 2, 1], allowing the model to capture coarse-to-fine temporal patterns from biweekly up to monthly scales. One block in each stack had two fully connected layers of 512 units with dropout (0.1). The hierarchical downsampling with subsequent interpolation enables N-HiTS to capture long-term trends and short-term oscillations jointly.

Neural Basis Expansion Analysis for Time Series (N-BEATS):

N-BEATS (Oreshkin et al., 2019) decomposes the predictions into understandable basis functions through three stacks dedicated to generic (learning data-driven basis), trend (polynomial basis), and seasonality (Fourier basis with 2 harmonics), respectively. The stack consisted of two blocks containing 64 hidden units each. Such a formulation mathematically separates trend and seasonal

patterns while allowing a generic stack to learn residual patterns, providing mechanistic interpretability of drought index temporal structure.

eXtreme Gradient Boosting (XGBoost):

Since the sole non-neural architecture, XGBoost constructs an ensemble of 300 decision trees with gradient boosting, depth 6, and learning rate 0.05. Regularization was achieved with L1 ($\alpha = 0.1$) and L2 ($\lambda = 1.0$) penalties, subsampling (0.8 rows and 0.8 features), and early stopping (20 rounds). The 12-month horizon was flattened into feature vectors, with each lag as a separate feature. The tree-based nature of XGBoost naturally has the natural treatment of non-linear relationships and feature interactions without requiring explicit architectural design for temporal dependencies. (Chen and Guestrin, 2016).

All of the model architectures are presented in Figure 4, with hyperparameters completely defined in the specification table given.

3.3.3 Training and Optimization

Model training was performed on a common protocol among architectures for fair comparison. The mean squared error (MSE) loss function was applied to all the neural network models because of the gradient properties of the loss function and agreement with accuracy metrics for the task of forecasting. For the PyTorch-based models (LSTM, TCN, N-HiTS, N-BEATS), we used the AdamW optimizer with weight decay (1×10^{-4}) and gradient clipping (max norm = 1.0) to prevent exploding gradients. (Pascanu et al., 2012). TensorFlow-based BiLSTM used the Adam optimizer. (Loshchilov and Hutter, 2017). The learning rates were the same for all neural models, namely, 1×10^{-3}, and the batch sizes varied from 64 (BiLSTM) to 256 (TCN) based on memory demand and convergence behavior.

Training was conducted for a maximum of 100-500 epochs depending on the complexity of the model, with the validation loss tracked through early stopping. Training was halted in case validation loss did not decrease in 10 consecutive iterations (patience = 10), and the best-performing weights were stored. This prevented overfitting but allowed sufficient training for convergence. Longer training (500 epochs for SPEI) was permitted for BiLSTM due to the strength of the model and long-lookback sequence complexity.

XGBoost training employed an alternate protocol more suitable for tree-based learning: 300 boosting iterations with early stopping after 20 non-improving rounds on the validation. Tree building used column and row subsampling (0.8 each) and minimum child weight (3) to control overfitting, and all available CPU cores were utilized for parallel tree building.

They were trained individually for each of the 36 prediction tasks (9 drought indexes × 4 ecosystems), enabling index-specific and ecosystem-specific parameterization. They were trained using NVIDIA GPU hardware for neural network models (CUDA-ready PyTorch 2.0+ and TensorFlow 2.x), and CPU-based training was conducted for XGBoost. Training time for each

model ranged from 2-5 minutes (BiLSTM-based GPU) to 10-15 minutes (XGBoost-based CPU), totaling overall computation of approximately 20 hours for all 216 models (36 tasks × 6 architectures).

All the computations were performed on a system with an NVIDIA GeForce RTX 4060 GPU (8GB VRAM), Intel Core i7-13700H CPU, and 32 GB RAM under Windows 11 with CUDA 12.1 GPU acceleration.

3.3.4 Performance Evaluation

Model performance was evaluated using three complementary metrics computed on the held-out 2024 test set. (Hyndman and Koehler, 2006). The coefficient of determination (R^2) quantifies the proportion of variance in observed values explained by predictions:

$$R^2 = 1 - \frac{\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2}{\Sigma_{i=1}^{n}(y_i - \bar{y})^2}$$

where y_i observed values, $\hat{y}_i$ predictions, $\bar{y}_i$ Is the observed mean, and nis the number of test samples. R^2 ranges from $-\infty$ to 1, with 1 indicating perfect prediction and negative values indicating performance worse than a naive mean forecast. R^2 was selected as the primary metric for model comparison due to its scale-invariant properties, enabling direct comparison across drought indices with different units and magnitudes.

Root mean squared error (RMSE) quantifies average prediction error in the original units of each drought index:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

RMSE provides the absolute scale of forecast accuracy in the same units as the target variable, making it simple to understand practical forecast skill.

For each task, forecasts were generated on all selected valid pixels within the 2024 test window, then spatially aggregated by computing the pixel-mean over the land type. Metrics were calculated on these spatially aggregated time series, which convey landscape-scale forecast skill relevant to regional drought management. This aggregation of methods reduces noise from single-pixel variability while maintaining temporal dynamics critical for operational forecasting. Model rankings were determined using R^2 as the primary criterion, with secondary confirmation of forecast accuracy using RMSE.

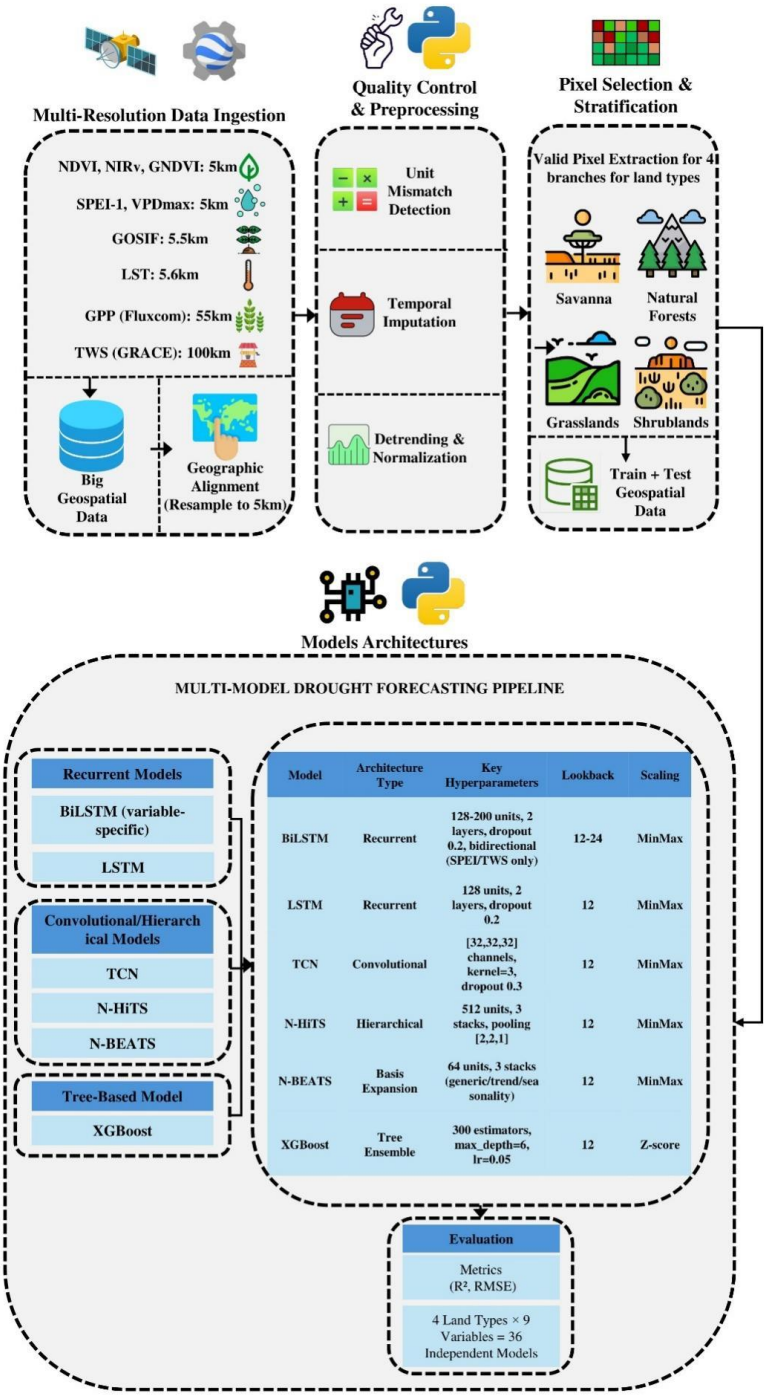


Figure 4. Multi-model drought forecasting approach. The pipeline includes four stages: (1) Multi-resolution ingestion of model-based and satellite data at native resolutions ranging from 5 km to 100 km, (2) Preprocessing and quality control including unit mismatch detection, temporal imputation for gap-filled LST and TWS, monthly detrending, and model-specific normalization, (3) Pixel selection and stratification of valid representative pixels for four natural land types

(Natural Forests, Savannas, Grasslands, Shrublands), and (4) Parallel training of six deep learning architectures divided by paradigm (recurrent, convolutional/hierarchical, and tree-based). All models perform 1-month ahead univariate forecasting on 36 independent tasks (9 drought indices × 4 land types). Training employed data between 2001 and 2021, validation between 2022 and 2023, and testing examined generalization against 2024 data. Model performance was quantified by R^2 and RMSE on spatially aggregated predictions.

4. Results

4.1 Multi-Sensor Validation and Inter-Index Relationships

Correlation analysis showed a strong agreement among vegetation productivity indicators. The indices GNDVI, GOSIF, GPP, NIRv, and NDVI were highly correlated (r = 0.82–0.98) across all land cover types, confirming that satellite observations reliably capture ecosystem functioning. Notably, GOSIF exhibited an excellent correlation with GPP derived from ground observations (r = 0.96–0.98), validating the accuracy of satellite-based productivity estimates (Figure 5).

Among greenness indices, NIRv performed better than the traditional NDVI in representing GPP (r = 0.62–0.82 versus 0.55–0.73), supporting its use as a more robust proxy for canopy productivity.

Meteorological and hydrological indicators showed relatively weak correlations with vegetation indices. SPEI and TWS were nearly independent (r = 0.09–0.22), indicating they represent distinct drought processes—meteorological versus hydrological. As expected, VPD correlated negatively with vegetation indices (r = –0.41 to –0.81), reflecting the inhibitory effect of high atmospheric demand on photosynthesis (Figure 5).

Cross-correlation analysis revealed that vegetation indices responded rapidly to environmental stress, with peak correlations occurring at lag = 0, indicating an immediate reaction to changing conditions. In contrast, TWS and SPEI exhibited notable memory effects, with distributed response lags of ±2–3 months, reflecting the delayed hydrological and meteorological adjustments within the system. Among ecosystem types, Natural Forests showed a delayed response to VPD, peaking at a +1-month lag, whereas Grasslands responded instantaneously (lag = 0), consistent with differences in rooting depth and water-use strategies (Figure 6).

These results suggest that vegetation responses to drought are closely tied to the dominant water acquisition and regulation strategies of different ecosystems. The immediate reaction of Grasslands to atmospheric stress indicates a shallow-rooted system with limited buffering capacity, making them more vulnerable to short-term fluctuations in vapor pressure deficit and soil moisture. Conversely, the delayed response observed in Natural Forests reflects deeper rooting systems and greater hydraulic capacitance, allowing trees to access sub-surface water reserves and maintain physiological activity despite transient atmospheric stress. The memory effects in TWS and SPEI further highlight that hydrological recovery lags behind vegetation greenness, implying that ecosystems can visually recover while subsurface water stores remain depleted. This temporal

decoupling between vegetation response and hydrological restoration underscores the importance of integrating multiple indicators when assessing drought resilience and ecosystem recovery.

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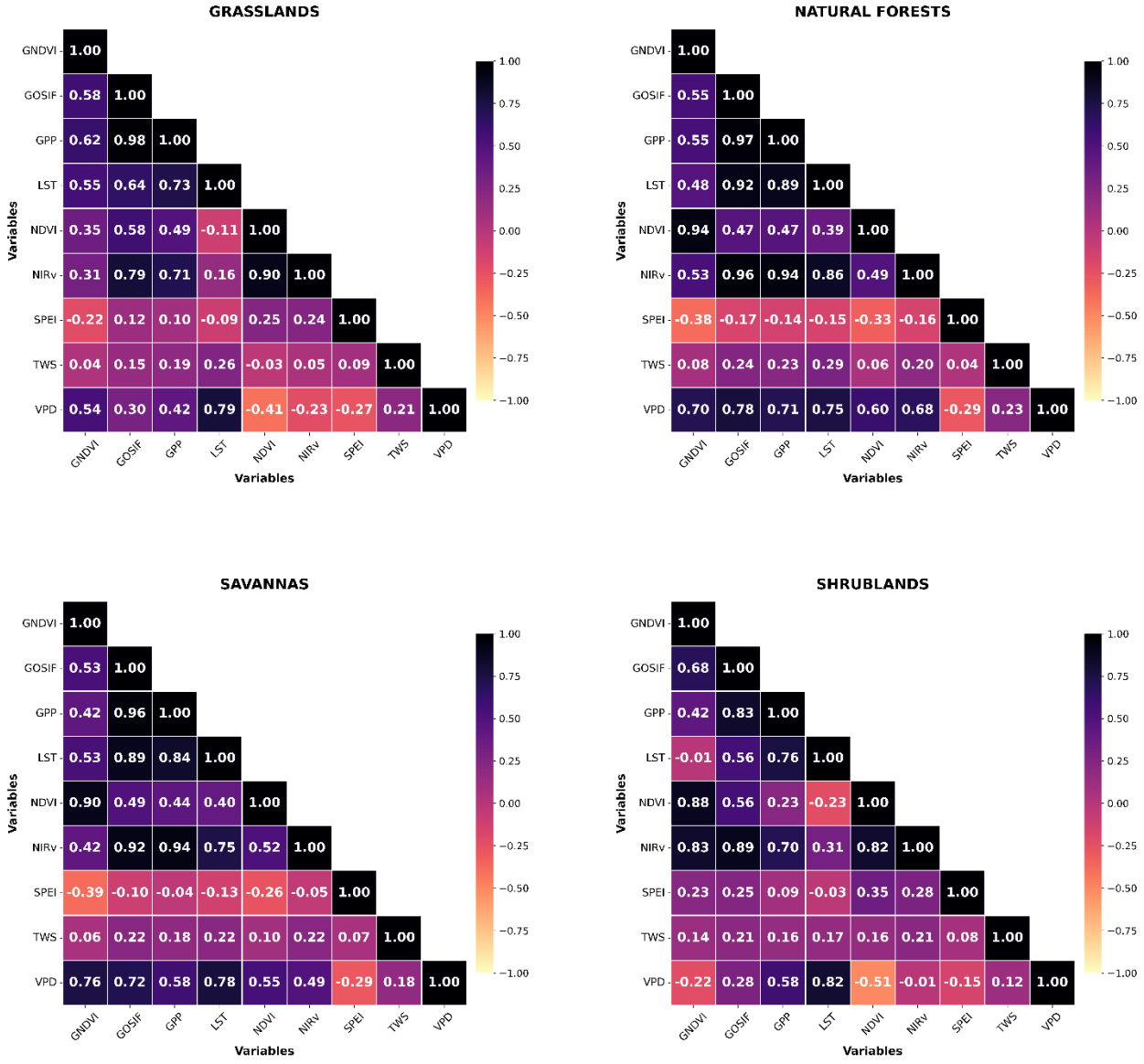


Figure 5. Inter-index correlation matrices (2001-2024) across four California land types showing relationships between nine drought indices.

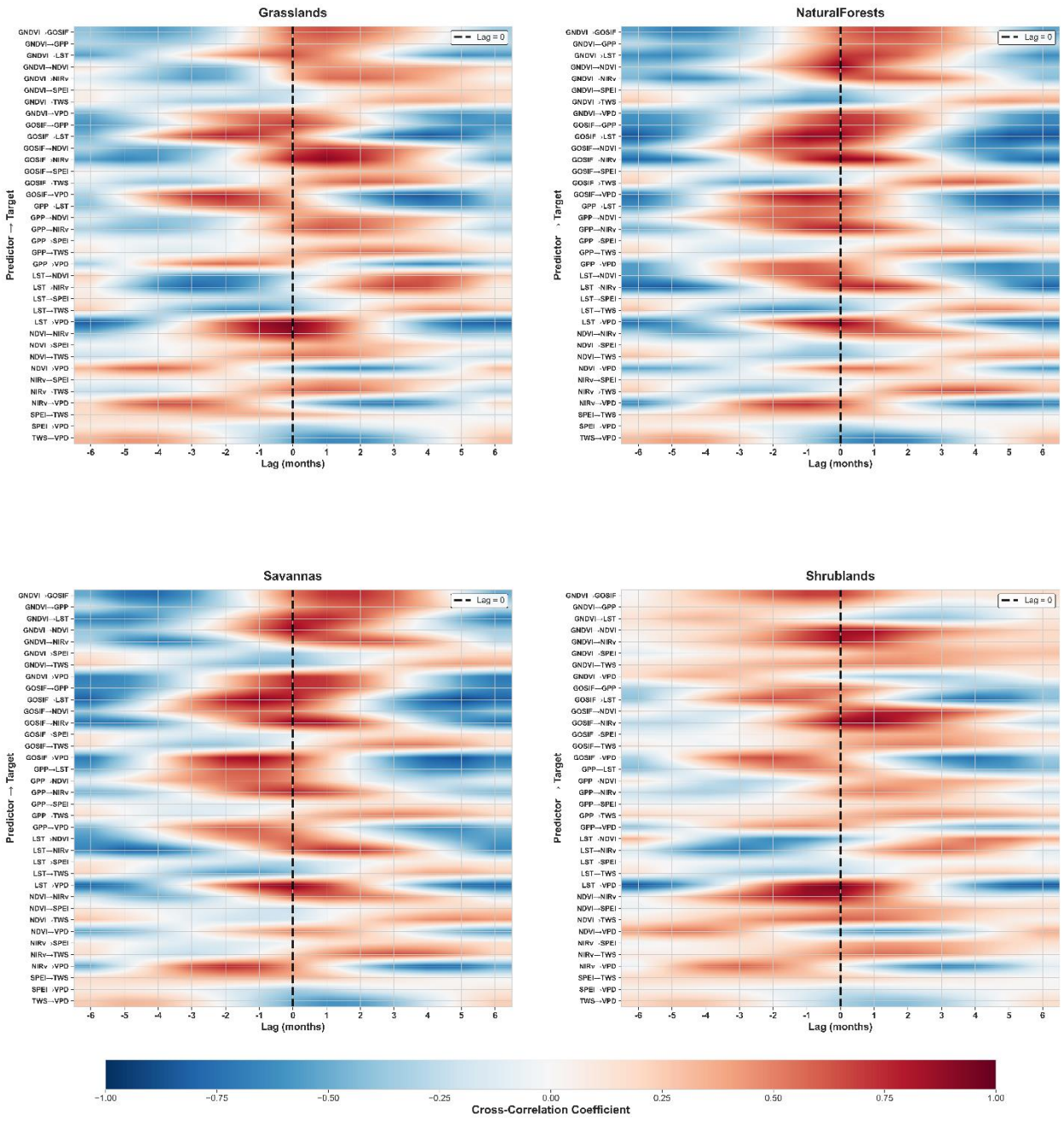


Figure 6. Cross-correlation functions showing temporal lag relationships between predictor-target pairs across land types. Lag-0 dominance for vegetation indices indicates rapid response; distributed patterns for TWS/SPEI suggest hydrological memory.

4.2 Deep Learning Forecasting Performance

Six deep learning models were evaluated across 216 forecasting tasks (9 indices × 4 land types × 6 models). Among them, BiLSTM demonstrated the most consistent and robust performance, achieving the highest coefficient of determination (R^2) in 47% of all cases and the lowest root mean square error (RMSE) in 94% of cases (Table 2).

584 **Table 2.** Model performance rankings across all forecasting tasks.

Model	Best R^2 Count	Best RMSE Count	Mean R^2	Mean RMSE
BiLSTM	17/36 (47%)	34/36 (94%)	0.62	0.06
TCN	13/36 (36%)	1/36 (3%)	0.60	0.58
N-HiTS	4/36 (11%)	1/36 (3%)	0.58	0.56
XGBoost	1/36 (3%)	0/36 (0%)	0.46	0.56
N-BEATS	1/36 (3%)	0/36 (0%)	0.55	0.57
LSTM	0/36 (0%)	0/36 (0%)	0.61	0.54

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586 Predictability differed markedly by drought indicator. Vegetation indices (GNDVI, NDVI, NIRv)
587 achieved the strongest performance ($R^2 > 0.80$) owing to their strong seasonal and phenological
588 regularity. Productivity and thermal indices (GOSIF, GPP, LST) showed moderate predictability
589 ($R^2 = 0.40–0.80$), reflecting both physiological and environmental influences. In contrast,
590 meteorological and hydrological indices (SPEI, TWS, VPD) were the most challenging to forecast
591 ($R^2 < 0.40$), with SPEI and VPD occasionally producing negative R^2 values, indicating
592 performance below that of a naïve climatological mean forecast (Table 3).

593 **Table 3.** Index-specific predictability and forecast difficulty.

Predictability	Index	Mean R^2	Best Model
Simple ($R^2 > 0.80$)	GNDVI	0.82	BiLSTM/XGBoost
	NDVI	0.87	BiLSTM/TCN
	NIRv	0.88	BiLSTM
	GOSIF	0.90	BiLSTM
Moderate (R^2 0.40-0.75)	LST	0.71	TCN
	VPD	0.41	BiLSTM
	GPP	0.56	BiLSTM
Challenging ($R^2 < 0.40$)	SPEI	0	BiLSTM
	TWS	0	BiLSTM

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For the more difficult indices, such as SPEI, BiLSTM slightly outperformed all other models (R^2 = 0.18 vs. TCN = 0.13). However, for vegetation-based indices with strong periodic behavior, Figure 7 shows that simpler architectures achieved comparable results, suggesting that model complexity should be proportional to temporal complexity in the data.

Across land cover types, forecast skill ranked as follows: Grasslands (mean R^2 = 0.67) > Savannas (0.57) > Natural Forests (0.56) > Shrublands (0.47), indicating that ecosystems with rapid phenological turnover are more predictable at monthly scales than those with slower physiological dynamics.

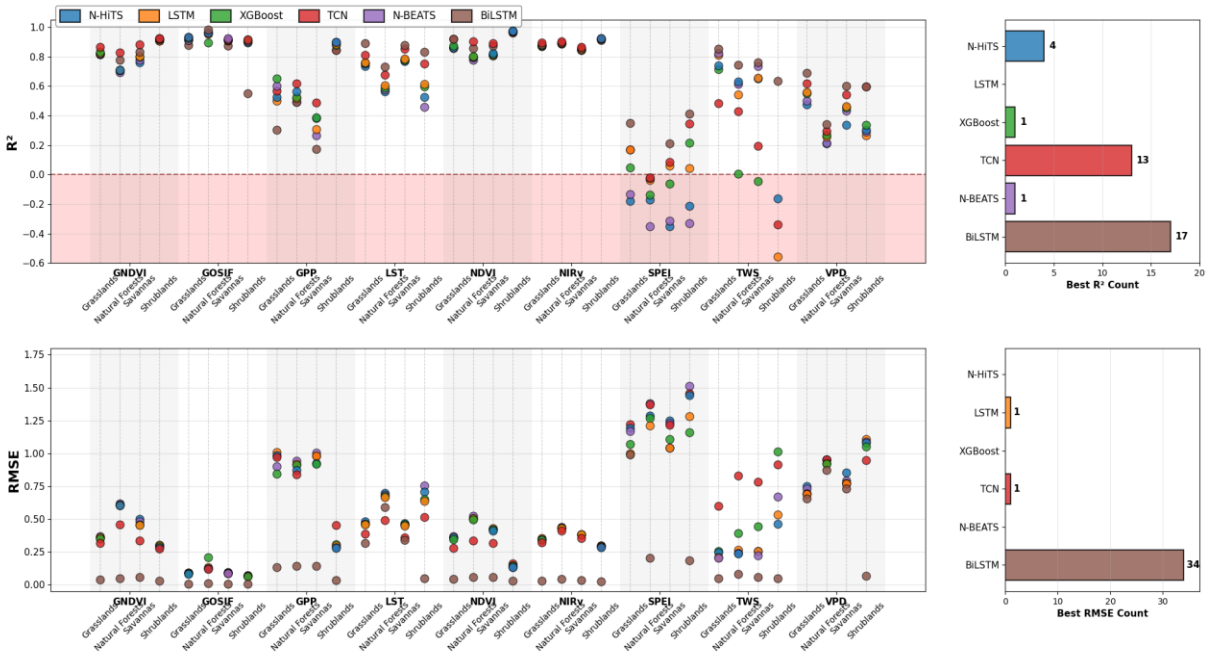


Figure 7. Forecasting performance (R^2 and RMSE) of six deep learning architectures across nine drought indices and four land types on held-out 2024 test data.

These results highlight that the predictability of drought-related processes depends strongly on the degree of ecosystem coupling to atmospheric variability. Vegetation-based indices, which encapsulate direct biophysical responses such as greenness and canopy structure, retain strong seasonal regularity that deep learning models can easily capture. In contrast, indices representing subsurface and atmospheric processes (e.g., TWS, VPD, SPEI) involve delayed or nonlinear feedbacks with vegetation and climate, posing challenges for one-month lead predictions. The superior performance of BiLSTM underscores the importance of architectures that preserve temporal memory in modeling drought evolution. These findings emphasize the value of integrating multi-sensor observations with temporal-aware models for advancing operational drought forecasting and ecosystem resilience monitoring across diverse Californian landscapes.

4.3 Drought Event Comparison: The Resilience Paradox

Comparison of NIRv responses established a paradox: whereas TWS remained at -2 to -3 SD beneath baseline in 2020-2024 with no recovery (Figure 3), vegetation responded differently. Natural Forests and Shrublands recovered strongly: Natural Forests from -8.9% to -0.8% (+8.1 points) and Shrublands from -12.2% to -1.7% (+10.5 points). But Grasslands and Savannas declined: Grasslands fell slightly from -2.3% to -2.6% (-0.3 points), whereas Savannas fell sharply from -1.7% to -7.4% (-5.7 points) (Table 4).

However, spatial analysis detected bimodal reactions: mean conditions for forests and shrublands did recover, yet severe stress zones (>40% decrease) rose significantly for all land cover types. Natural Forests experienced the largest increase in severe stress, 17 times, from 0.14% to 2.42% forest cover. Savannas experienced 14 times expansion (0.11% to 1.57%), and it brought oak woodland-grassland ecotones to the forefront of conservation interests. Even in more favorable land classes, areas of vulnerability expanded: Grasslands severe stress increased 4.6 times (0.38% to 1.75%), and Shrublands increased 2 times (0.40% to 0.81%).

Table 4. Mean NIRv response and severe-stress area expansion across drought periods.

Land Type	2012-2016 Mean (%)	2020-2024 Mean (%)	Improvement (pts)	Severe Stress 2012-2016 (%)	Severe Stress 2020-2024 (%)	Expansion Factor
Grasslands	-2.3	-2.6	-0.3*	0.38	1.75	4.6×
Natural Forests	-8.9	-0.8	+8.1	0.14	2.42	17×
Savannas	-1.7	-7.4	-5.7*	0.11	1.57	14×
Shrublands	-12.2	-1.7	+10.5	0.40	0.81	2.0×

* Grasslands and Savannas showed deteriorating mean conditions despite overall improvements in other land types.

Persistent hotspots of susceptibility—pixels that declined >40% during both droughts—represented 0.02-0.33% of each type of land cover and had high spatial clustering (Moran's $I = 0.28\text{-}0.34$, $p < 0.001$). These repeating failure hotspots most likely reflect some species with poor adaptive capacity, shallow soils, or additive impacts from past disturbances (Figure 8).

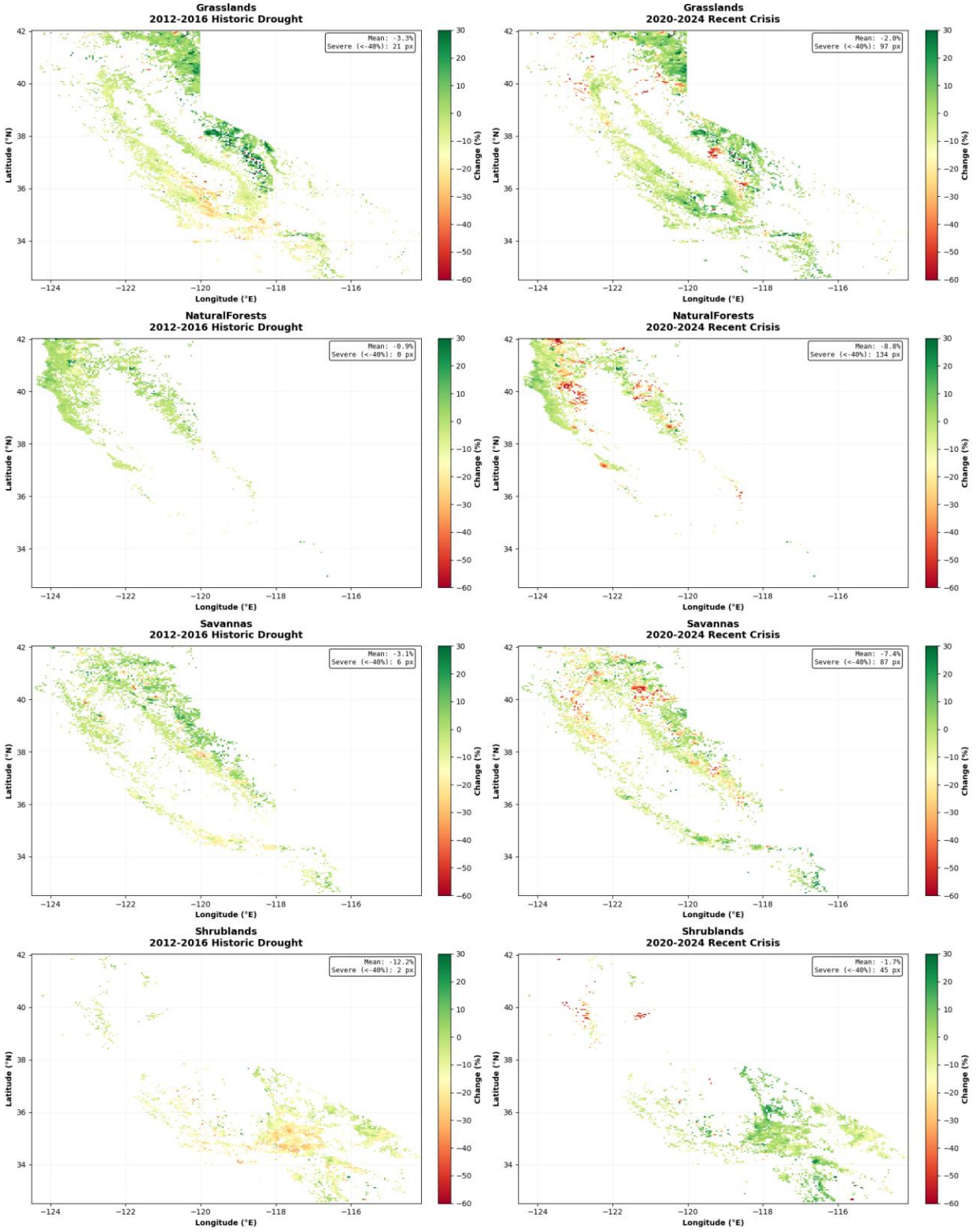


Figure 8. Spatial comparison of NIRv response during 2012-2016 historic EWBI drought vs. 2020-2024 recent period, relative to 2001-2011 baseline. Red = vegetation decline, green =

stable/recovery. Despite persistent TWS deficits, 2020-2024 shows substantially less stress but increased spatial heterogeneity.

4.4 Ecosystem State Transitions

Figure 9 illustrates EWBI' (stress) vs. NIRv' (response) state allocation indicated that the California ecosystems were in stressed on average of (31.2%) and healthy (28.5%) states for nearly equal durations between 2001 and 2024. Shrublands, which contain "stress traps" in which extremely stressed chaparral and coastal sage have a mere 26% monthly probability of recovery, have the maximum Dry/Stressed persistence (74%). Natural Forests had the lowest Wet/Healthy persistence (48.1%) and the highest Wet/Suppressed transitions (from Wet/Healthy state, 11.1% transition to Wet/Suppressed; Wet/Suppressed persistence at 50.0%), reflecting slow tree responses to alterations in water conditions where greater water availability is not instantaneously boosted to greater productivity.

Asymmetric transitions revealed that stress onset exceeded recovery: transitions from Wet/Healthy to Dry/Stressed (2.9-9.2%) exceeded transitions from Dry/Stressed to Wet/Healthy (1.4-6.4%) across all land types.

5. Discussion

5.1 The Paradox of Drought Resilience: Mechanisms and Impacts

Our most robust finding is the paradoxical decoupling of hydrological forcing and ecosystem response, 2020-2024. Faced with TWS never rebounding from 2-3 SD beneath baseline through 2024 and not improving—indicating chronic serious water deficits—plant stress showed divergent responses. Natural Forests and Shrublands exhibited substantial recovery (+8.1 and +10.5 percentage points, respectively), while Grasslands and Savannas deteriorated (-0.3 and -5.7 points). This decoupling of water supply and vegetation response negates the assumption of linear drought-ecosystem links and suggests fundamental changes in California's natural ecosystems.

We suggest three non-mutually-exclusive mechanisms for this paradox:

Ecological filtering and adaptation: The 2012-2016 drought would have likely caused drought-intolerant genotypes to perish, leaving behind surviving populations enriched in stress tolerance. Laboratory experiments in California forests following the 2012-2016 drought recorded 129 million dead trees that were localized within species with conservative water-use strategies. (Asner et al., 2016; Restaino et al., 2019). Surviving individuals might be a filtered group with larger root systems, anisohydric stomatal function, or enhanced cavitation resistance, facilitating continued function at severe water deficits.

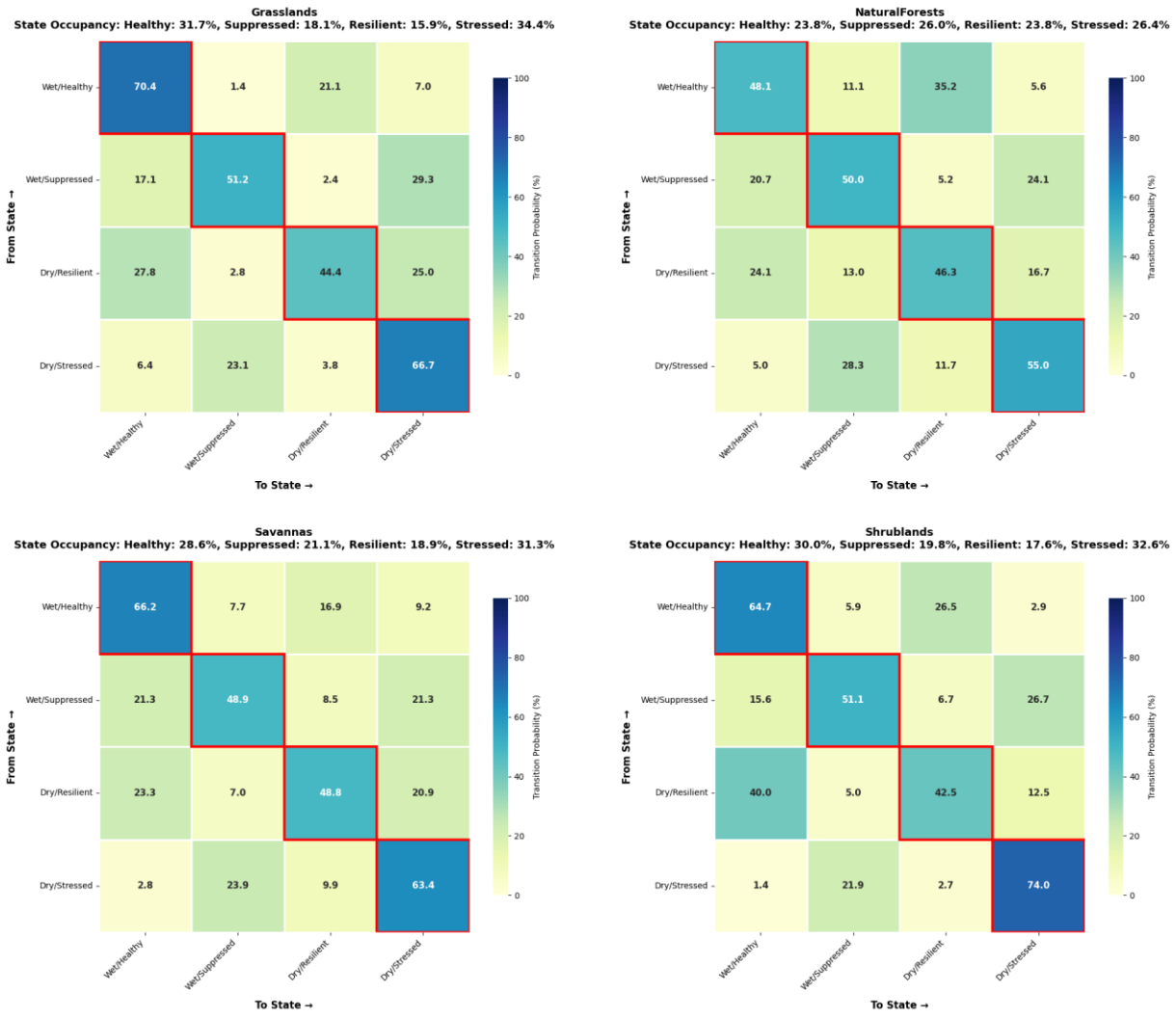


Figure 9. Ecosystem state transition matrices (2001-2024) showing monthly persistence and transition probabilities.

Differential drought characteristics: While both events had extreme TWS deficits, atmospheric demand characteristics were distinct. The 2012-2016 drought superimposed peak water deficiency on record VPD and temperatures. (Diffenbaugh et al., 2015). By contrast, 2020-2024 had more moderate VPD despite comparable TWS deficits (Figure 3), potentially alleviating evaporative stress and enabling ecosystems to maintain function with limited access to water. This highlights that absolute water availability is not in itself a good predictor of vegetation response—atmospheric demand co-limits ecosystem function.

Groundwater access stratification: Environmental systems with deep-access to groundwater storage can be disentangled from TWS surface signals. 20 km³ groundwater loss in California Central Valley occurred from 2012-2016, and water table declines of 30-60 m in strongly pumped regions (Xiao et al., 2017). Deep-rooted vegetation with access to stable groundwater would

exhibit resistance irrespective of TWS deficits, reflecting primarily surface/shallow soil water loss. This process should create spatial heterogeneity related to aquifer depth and recharge areas.

5.2 Bimodal Responses of the Ecosystem and Hotspots of Vulnerability

While mean conditions recovered for some land types, areas of extreme stress rose 2-17 fold, laying bare bifurcation into healthy and crashing subpopulations. Natural Forests diverged most dramatically: mean stress improved 8.1 percentage points, but the area of extreme stress rose 17-fold from 0.14% to 2.42% of forestland. The bimodal response indicates that California's ecosystems are not responding uniformly but diverging in various paths.

Recurring hotspots of vulnerability—pixels losing >40% in both droughts—also showed strong spatial clustering (Moran's I = 0.28-0.34), implying these are not random failures but replicable reactions to specific conditions. These areas, we speculate, are:

Species-level constraints: Certain taxa lack physiological characteristics that will enable adaptation. Vegetation in California ranges with species along the isohydric-anisohydric axis; isohydric species close stomata early to prevent hydraulic failure, achieving carbon gain but at a cost of carbon starvation during prolonged drought. (McDowell et al., 2008).

Edaphic limitations: Impermeable lower horizons, shallow soil, or low water-holding capacity restrict penetration to deeper water by roots. Serpentine soils in California cover ~1% of the state and impose severe water and nutrient limitation on drought-prone endemic-rich communities (Harrison et al., 2004).

Cumulative decline: Sites disturbed repeatedly (fire, insect epizootic, prior drought) may be lacking resilience to future stress. Bark beetle epidemics 2012-2016 killed >102 million trees, with mortality concentrated in water-stressed stands before the epidemics. (Young et al., 2017). These stressed sites lack carbon reserves and intact hydraulic architecture necessary for drought recovery.

Extreme stress area expansion in the presence of improving means (for some land types) is of fundamental consequence. Even though regional averages can demonstrate resilience, the 2.42% of the Natural Forest area in extreme decline contains sufficient mortality extent to make regional carbon balances shift from sinks to sources. (Anderegg et al., 2020). Collapse zones are capable of triggering positive feedbacks: dead biomass increases fire hazard, decreases local moisture recycling, and creates microclimates that advance invasive species—potentially triggering larger-scale regime shifts. (Abatzoglou and Williams, 2016).

5.3 Ecosystem State Dynamics and Stress Traps

State transition analysis revealed underlying asymmetries in ecosystem dynamics. Shrublands experienced 74% Dry/Stressed persistence, meaning that once under high stress, chaparral and coastal sage have only 26% monthly possibility of recovery. For 12-month time scales, this implies a <1% possibility of escaping to the Wet/Healthy state without intervention. This "stress trap" reflects the long recovery times required for woody vegetation following severe drought damage.

Asymmetry of stress onset-recovery (transitions to Dry/Stressed are 1.5-6.6× more numerous than reverse transitions) is a manifestation of inherent thermodynamic constraints. Physiological damage occurs rapidly—cavitation propagates through xylem within hours to days under severe water stress—while repair requires energy-requiring processes with time scales of months to years: refilling embolized vessels, replenishing carbohydrate reserves, replacing damaged tissue. (Brodribb and Cochard, 2009). This hysteresis implies that brief wet spells occurring during multi-year droughts provide minimal opportunity for recovery, and extended favorable conditions are required for restoration.

Low Wet/Healthy persistence (48.1%) and frequent transitions to Wet/Suppressed state (11.1% monthly transition rate from Wet/Healthy; 50% Wet/Suppressed persistence) for Natural Forests are suggestive of physiological time lags in trees. Increased water availability must trigger seasonal developmental programs—bud break, leaf expansion, cambial reactivation—with 2 to 6-month time scales. Forests under California's brief wet seasons can experience increased water status without time for full photosynthetic upregulation before the next dry season onset.

The asymmetric transition dynamics imply decreasing marginal effectiveness of precipitation with increasing cumulative deficits. This non-linearity is in agreement with threshold models, where ecosystems remain resilient within a "safe operating space" but cross tipping points under chronic stress. (Scheffer et al., 2001). For management, this implies that early intervention is preferable to reactive management following the shift of ecosystems into locked degradation states.

5.4 Operational Design and Forecasting Performance

Model comparison using structured analysis indicated that forecast capability depends primarily on temporal complexity and not on land type or spatial features. BiLSTM achieved optimal RMSE in 94% of the tasks (34 of 36), demonstrating clear superiority in minimizing error. However, TCN achieved optimal R^2 most frequently (36%, 13 of 36 tasks), indicating that different architectures optimize different performance measures.

The hierarchy of predictability unique to each index—vegetation indices ($R^2 > 0.80$) vs. meteorological-hydrological indices ($R^2 < 0.40$)—reflects fundamental contrasts in temporal dynamics. Vegetation indices exhibit strong seasonal autocorrelation governed by phenology that makes reliable prediction from historical trends alone feasible. SPEI and TWS, on the other hand, integrate complex precipitation-infiltration-recharge processes with stochastic forcing into low-frequency variability that is weakly predictable from recent history.

For operational forecasting systems, these findings suggest matching model complexity to the forecast target. When forecasting vegetation indices with strong seasonality, simpler models (XGBoost, simple LSTM) achieve comparable performance (R^2 within 5% of BiLSTM) with 10-20× faster inference. Critical early warning indices like SPEI and TWS require long lookback windows and bidirectional processing of BiLSTM to capture multi-scale temporal dependencies, where computational costs are justified.

Land-type forecast skill (Grasslands > Savannas > Natural Forests > Shrublands) agrees with structural complexity and response timescales. The high turnover (yearly lifecycles) of Grasslands fosters close coupling between environmental forcing and vegetation response, enhancing predictability. Natural Forests integrate signals over longer timescales through stored reserves and deep roots, including memory effects that render short-term forecasting more difficult.

The limited GPP forecast skill ($R^2 = 0.56$) is likely caused by the direct FLUXCOM-MOD17 switch inducing systematic methodological discrepancies (data-driven upscaling vs. process modeling). Future applications must address the provision of continuous data streams for long-term forecasting or apply transfer learning to handle data source changes.

5.5 Broader Context and Mediterranean Ecosystems

The California example provides a critical preview of the dynamics to be anticipated in Mediterranean-climate ecosystems worldwide. Such ecosystems—the Mediterranean Basin, central Chile, southwestern Australia, the Cape region of South Africa—share convergent features: high species richness, summer drought adaptation, fire-prone landscapes, and predicted intensification of drought under climate change (Diffenbaugh and Field, 2013). Our findings have the following immediate relevance:

- *Potential for adaptation is present but spatially variable.* While Natural Forests and Shrublands showed improved mean conditions in 2020-2024, a 17-fold expansion of severe stress areas indicates limits are being approached for vulnerable subpopulations. The evolutionary history of Mediterranean ecosystems with seasonal drought does not necessarily mean resilience to multi-year droughts outside historical variability (Allen et al., 2015).
- *Spatial heterogeneity requires high-resolution tracking*. Regional mean measures conceal critical spatial variability in ecosystem responses. While average conditions may improve for some land types, simultaneously expanding collapse zones reveal emerging vulnerability hotspots that require targeted intervention. Conservation must move beyond landscape-scale examination to find and protect vulnerable microsites.
- *Hysteresis complicates recovery.* The asymmetry between rapid stress onset and slow recovery means that drought impacts persist long after meteorological conditions improve. Management interventions—competition reduction, assisted migration, prescribed fire to reduce ladder fuels—may be required to escape locked degradation states.
- *Multi-sensor integration is necessary.* Single-drought index use misses important dimensions. Our EWBI composite integrates water scarcity (SPEI, TWS) and atmospheric/thermal stress (VPD, LST) synergistic impacts collectively determining ecosystem response. Monitoring systems in operational applications need to integrate multiple dimensions to avoid false negatives (perceived water availability masking atmospheric stress) or false positives (overestimated atmospheric stress masking vegetation response when groundwater is available).

5.6 Limitations and Future Directions

Several caveats are worth noting:

- Our 2001-2011 baseline may not accurately represent "natural" conditions—California experienced a record drought in 2007-2009, possibly biasing baseline toward already-stressed states. Analysis with longer time frames using pre-2000 conditions would strengthen inference into ecosystem trajectories.
- Five km spatial resolution, while finer than most published drought research, is still coarse enough to suppress significant heterogeneity. Field validation linking remote sensing signals to plot-scale composition, demographics, and physiological status would ground-truth our hotspot detections for vulnerability.
- Our single-variable forecasting model addresses each index separately, regardless of possible cross-variable correlations. Multivariate models forecasting vegetation based on meteorological-hydrological forcing would enhance forecast performance and yield a mechanistic interpretation of drought effects.
- The FLUXCOM-MOD17 GPP transition introduces uncertainty to estimates of productivity. While both products estimate ecosystem carbon uptake, methodology differences limit confidence in temporal trends, including the transition. Ensemble methods that combine several GPP products to estimate uncertainty must be explored in future research.
- Our analysis is restricted to natural systems, leaving out the 25% of California land cover devoted to agriculture or urban uses. Managed systems differ from one another in water availability (irrigation), stress behavior (crop phenology), and management practices (fallowing, groundwater pumping), producing a different drought dynamic needing a separate examination.

Future research agendas include: (1) field experiments validating vulnerable pixel characteristics through species surveys, soil measurements, and dendrometer data; (2) long-term temporal examination through pre-2000 conditions and projections to put recent droughts into a centennial perspective; (3) multivariate predictive models with meteorological forcing to predict vegetation responses; (4) comparative studies among Mediterranean-climate regions to test generality of observed dynamics; and (5) experimental manipulations to test management interventions (thinning, prescribed fire, assisted migration) within areas of identified vulnerability.

6. Conclusion

This paper presents an integrated framework that includes multi-sensor drought monitoring, deep learning prediction, and ecosystem state dynamics to characterize resilience patterns and predictability in California natural landscapes. Three key findings are unveiled:

First, California ecosystems displayed paradoxical and spatially divergent responses during 2020-2024. Natural Forests and Shrublands showed substantial mean improvements (+8.1 and +10.5 percentage points, respectively) compared to the 2012-2016 drought, while Grasslands and Savannas deteriorated (-0.3 and -5.7 points), despite persistent extreme TWS deficits remaining at -2 to -3 SD below baseline through 2024. This decoupling can plausibly be explained by ecological filtering (drought tolerance-selective mortality), differential atmospheric demand (reduced VPD during 2020-2024), and stratified groundwater availability. However, the mean resilience for improving land types conceals vital spatial heterogeneity.

Second, extreme stress locations increased 2 to 17-fold across land types despite improved means for some, diverging bimodally into collapsing and resistant subpopulations. Natural Forests showed 17-fold expansion (0.14% → 2.42% of forest cover), sufficient to alter regional carbon balances. Persistent hotspots of weakness with repeated failure during both droughts comprise 0.02-0.33% of each land unit and have high spatial clustering (Moran's I = 0.28-0.34), indicating replicable constraints due to species attributes, edaphic limitations, or cumulative degradation. These sites require targeted field investigation and management intervention.

Third, analysis of ecosystem state showed asymmetric transition dynamics with stress onset being 1.5-6.6× more frequent than recovery. Shrublands showed 74% Dry/Stressed persistence reflecting stress traps, and all land types demonstrated hysteresis requiring persistent favorable conditions for recovery as opposed to short wet periods. This asymmetry reflects fundamental thermodynamic constraints where physiological damage occurs rapidly, but repair requires months to years.

For operational forecasting, BiLSTM achieved the best RMSE performance (94% best, 34 of 36 tasks), and TCN achieved the best R^2 most frequently (36%, 13 of 36 tasks). Index predictability ranged from simple (vegetation: $R^2 > 0.80$) to challenging (SPEI, TWS, VPD: $R^2 < 0.40$) and indicates model complexity needs to be aligned with temporal dynamics. Grasslands had the highest predictive ability ($R^2 = 0.67$), Shrublands had the lowest ($R^2 = 0.47$), consistent with structural complexity and response timescales.

This research advances understanding of drought resilience in Mediterranean ecosystems by making the following identification: (1) repeated drought causes ecosystem divergence rather than homogeneous decline; (2) landscape-scale approaches conceal deep spatial remodeling with critical management implications; (3) hysteresis in ecosystem dynamics makes recovery difficult and may require active management; and (4) multi-sensor fusion tracks drought aspects single indexes cannot capture.

For California and Mediterranean-type regions globally, this study indicates that adaptation potential exists, but approaching thresholds are emerging in expanding collapse areas. Conservation interventions have to identify and protect hotspots of vulnerability before degradation irreversibility sets in. Preventive action is better than reactive strategies once ecosystems are trapped in states of stress. Multi-dimensional monitoring integrating water availability, atmospheric demand, and vegetation response will give effective early warning.

The methodology developed here—intertwining retrospective resilience analysis with operational predictive capability—offers a replicable template for other locations seeking to enhance climate resilience through evidence-based ecosystem management.

Acknowledgment

The author gratefully acknowledges Dr. Kanishka Mallick Raghuvanshi for his exceptional guidance, insightful feedback, and sustained support throughout this research. His expertise and mentorship were indispensable at every stage—from conceptual development to methodological refinement and interpretation of results—and this work would not have been possible without his contribution.

Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request. The code used for data processing, analysis, and figure generation is publicly available at: https://github.com/Aria2133/Linking-Drought-with-Vegetation-Response

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve language clarity and structure. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Abatzoglou, J.T., 2013. Development of gridded surface meteorological data for ecological applications and modelling. Intl Journal of Climatology 33, 121–131. https://doi.org/10.1002/joc.3413

Abatzoglou, J.T., Williams, A.P., 2016. Impact of anthropogenic climate change on wildfire across western US forests. Proc. Natl. Acad. Sci. U.S.A. 113, 11770–11775. https://doi.org/10.1073/pnas.1607171113

AghaKouchak, A., Farahmand, A., Melton, F.S., Teixeira, J., Anderson, M.C., Wardlow, B.D., Hain, C.R., 2015. Remote sensing of drought: Progress, challenges and opportunities. Reviews of Geophysics 53, 452–480. https://doi.org/10.1002/2014RG000456

Allen, C.D., Breshears, D.D., McDowell, N.G., 2015. On underestimation of global vulnerability to tree mortality and forest die-off from hotter drought in the Anthropocene. Ecosphere 6, 1–55. https://doi.org/10.1890/ES15-00203.1

Anderegg, W.R.L., Klein, T., Bartlett, M., Sack, L., Pellegrini, A.F.A., Choat, B., Jansen, S., 2016. Meta-analysis reveals that hydraulic traits explain cross-species patterns of drought-

induced tree mortality across the globe. Proc. Natl. Acad. Sci. U.S.A. 113, 5024–5029. https://doi.org/10.1073/pnas.1525678113

Anderegg, W.R.L., Trugman, A.T., Badgley, G., Anderson, C.M., Bartuska, A., Ciais, P., Cullenward, D., Field, C.B., Freeman, J., Goetz, S.J., Hicke, J.A., Huntzinger, D., Jackson, R.B., Nickerson, J., Pacala, S., Randerson, J.T., 2020. Climate-driven risks to the climate mitigation potential of forests. Science 368, eaaz7005. https://doi.org/10.1126/science.aaz7005

Asner, G.P., Brodrick, P.G., Anderson, C.B., Vaughn, N., Knapp, D.E., Martin, R.E., 2016. Progressive forest canopy water loss during the 2012–2015 California drought. Proc. Natl. Acad. Sci. U.S.A. 113. https://doi.org/10.1073/pnas.1523397113

Badgley, G., Field, C.B., Berry, J.A., 2017. Canopy near-infrared reflectance and terrestrial photosynthesis. Sci. Adv. 3, e1602244. https://doi.org/10.1126/sciadv.1602244

Bai, S., Kolter, J.Z., Koltun, V., 2018. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. https://doi.org/10.48550/ARXIV.1803.01271

Brodribb, T.J., Cochard, H., 2009. Hydraulic Failure Defines the Recovery and Point of Death in Water-Stressed Conifers. Plant Physiology 149, 575–584. https://doi.org/10.1104/pp.108.129783

Challu, C., Olivares, K.G., Oreshkin, B.N., Garza Ramirez, F., Mergenthaler Canseco, M., Dubrawski, A., 2023. NHITS: Neural Hierarchical Interpolation for Time Series Forecasting. AAAI 37, 6989–6997. https://doi.org/10.1609/aaai.v37i6.25854

Chen, L., Lin, Z., Li, H., Liu, X., 2025. Drought duration outweighs intensity in governing ecosystem recovery time. Science of The Total Environment 1009, 181056. https://doi.org/10.1016/j.scitotenv.2025.181056

Chen, T., Guestrin, C., 2016. XGBoost: A Scalable Tree Boosting System, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Presented at the KDD '16: The 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM, San Francisco California USA, pp. 785–794. https://doi.org/10.1145/2939672.2939785

Chornesky, E.A., Ackerly, D.D., Beier, P., Davis, F.W., Flint, L.E., Lawler, J.J., Moyle, P.B., Moritz, M.A., Scoonover, M., Byrd, K., Alvarez, P., Heller, N.E., Micheli, E.R., Weiss, S.B., 2015. Adapting California's Ecosystems to a Changing Climate. BioScience 65, 247–262. https://doi.org/10.1093/biosci/biu233

Cushman, S.A., McGarigal, K., 2019. Metrics and Models for Quantifying Ecological Resilience at Landscape Scales. Front. Ecol. Evol. 7, 440. https://doi.org/10.3389/fevo.2019.00440

Didan, K., 2021. MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061. https://doi.org/10.5067/MODIS/MOD13Q1.061

Diffenbaugh, N.S., Field, C.B., 2013. Changes in Ecologically Critical Terrestrial Climate Conditions. Science 341, 486–492. https://doi.org/10.1126/science.1237123

Diffenbaugh, N.S., Swain, D.L., Touma, D., 2015. Anthropogenic warming has increased drought risk in California. Proc. Natl. Acad. Sci. U.S.A. 112, 3931–3936. https://doi.org/10.1073/pnas.1422385112

Dikshit, A., Pradhan, B., Alamri, A.M., 2021. Long lead time drought forecasting using lagged climate variables and a stacked long short-term memory model. Science of The Total Environment 755, 142638. https://doi.org/10.1016/j.scitotenv.2020.142638

Elbeltagi, A., Srivastava, A., Ehsan, M., Sharma, G., Yu, J., Khadke, L., Gautam, V.K., Awad, A., Jinsong, D., 2024. Advanced stacked integration method for forecasting long-term drought severity: CNN with machine learning models. Journal of Hydrology: Regional Studies 53, 101759. https://doi.org/10.1016/j.ejrh.2024.101759

Friedl, M., Sulla-Menashe, D., 2022a. MODIS/Terra+Aqua Land Cover Type Yearly L3 Global 500m SIN Grid V061. https://doi.org/10.5067/MODIS/MCD12Q1.061

Friedl, M., Sulla-Menashe, D., 2022b. MODIS/Terra+Aqua Land Cover Type Yearly L3 Global 500m SIN Grid V061. https://doi.org/10.5067/MODIS/MCD12Q1.061

Gitelson, A.A., Kaufman, Y.J., Merzlyak, M.N., 1996. Use of a green channel in remote sensing of global vegetation from EOS-MODIS. Remote Sensing of Environment 58, 289–298. https://doi.org/10.1016/S0034-4257(96)00072-7

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment 202, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Graves, A., Fernández, S., Schmidhuber, J., 2005. Bidirectional LSTM Networks for Improved Phoneme Classification and Recognition, in: Duch, W., Kacprzyk, J., Oja, E., Zadrożny, S. (Eds.), Artificial Neural Networks: Formal Models and Their Applications – ICANN 2005, Lecture Notes in Computer Science. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 799–804. https://doi.org/10.1007/11550907_126

Greene, C., 2018. Broadening understandings of drought – The climate vulnerability of farmworkers and rural communities in California (USA). Environmental Science & Policy 89, 283–291. https://doi.org/10.1016/j.envsci.2018.08.002

Hao, Z., Singh, V.P., Xia, Y., 2018. Seasonal Drought Prediction: Advances, Challenges, and Future Prospects. Reviews of Geophysics 56, 108–141. https://doi.org/10.1002/2016RG000549

Harrison, S., Safford, H., Wakabayashi, J., 2004. Does the Age of Exposure of Serpentine Explain Variation in Endemic Plant Diversity in California? International Geology Review 46, 235–242. https://doi.org/10.2747/0020-6814.46.3.235

Hochreiter, S., Schmidhuber, J., 1997. Long Short-Term Memory. Neural Computation 9, 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Hulley, G., Hook, S., 2021. MODIS/Aqua Land Surface Temperature/3-Band Emissivity Daily L3 Global 0.05Deg CMG V061. https://doi.org/10.5067/MODIS/MYD21C1.061

Hyndman, R.J., Koehler, A.B., 2006. Another look at measures of forecast accuracy. International Journal of Forecasting 22, 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

Jiao, W., Wang, L., McCabe, M.F., 2021a. Multi-sensor remote sensing for drought characterization: current status, opportunities and a roadmap for the future. Remote Sensing of Environment 256, 112313. https://doi.org/10.1016/j.rse.2021.112313

Jiao, W., Wang, L., McCabe, M.F., 2021b. Multi-sensor remote sensing for drought characterization: current status, opportunities and a roadmap for the future. Remote Sensing of Environment 256, 112313. https://doi.org/10.1016/j.rse.2021.112313

Joiner, J., Yoshida, Y., 2020. Satellite-based reflectances capture large fraction of variability in global gross primary production (GPP) at weekly time scales. Agricultural and Forest Meteorology 291, 108092. https://doi.org/10.1016/j.agrformet.2020.108092

Jung, M., Schwalm, C., Migliavacca, M., Walther, S., Camps-Valls, G., Koirala, S., Anthoni, P., Besnard, S., Bodesheim, P., Carvalhais, N., Chevallier, F., Gans, F., Goll, D.S., Haverd, V., Köhler, P., Ichii, K., Jain, A.K., Liu, J., Lombardozzi, D., Nabel, J.E.M.S., Nelson, J.A., O'Sullivan, M., Pallandt, M., Papale, D., Peters, W., Pongratz, J., Rödenbeck, C., Sitch, S., Tramontana, G., Walker, A., Weber, U., Reichstein, M., 2020. Scaling carbon fluxes from eddy covariance sites to globe: synthesis and evaluation of the FLUXCOM approach. Biogeosciences 17, 1343–1365. https://doi.org/10.5194/bg-17-1343-2020

Kingma, D.P., Ba, J., 2014. Adam: A Method for Stochastic Optimization. https://doi.org/10.48550/ARXIV.1412.6980

Kleist, N.J., Domschke, C.T., Knight, A.C., Nauman, T.W., Duniway, M.C., Carter, S.K., 2025. Mapping predicted ecological states at landscape scales using remote-sensing data and machine learning. Ecosphere 16, e70243. https://doi.org/10.1002/ecs2.70243

Landerer, F.W., Flechtner, F.M., Save, H., Webb, F.H., Bandikova, T., Bertiger, W.I., Bettadpur, S.V., Byun, S.H., Dahle, C., Dobslaw, H., Fahnestock, E., Harvey, N., Kang, Z., Kruizinga, G.L.H., Loomis, B.D., McCullough, C., Murböck, M., Nagel, P., Paik, M., Pie, N., Poole, S., Strekalov, D., Tamisiea, M.E., Wang, F., Watkins, M.M., Wen, H., Wiese, D.N., Yuan, D., 2020. Extending the Global Mass Change Data Record: GRACE Follow-On Instrument and Science Data Performance. Geophysical Research Letters 47, e2020GL088306. https://doi.org/10.1029/2020GL088306

Lara-Benítez, P., Carranza-García, M., Riquelme, J.C., 2021. An Experimental Review on Deep Learning Architectures for Time Series Forecasting. Int. J. Neur. Syst. 31, 2130001. https://doi.org/10.1142/S0129065721300011

Lenihan, J.M., Drapek, R., Bachelet, D., Neilson, R.P., 2003. CLIMATE CHANGE EFFECTS ON VEGETATION DISTRIBUTION, CARBON, AND FIRE IN CALIFORNIA. Ecological Applications 13, 1667–1681. https://doi.org/10.1890/025295

Li, X., Xiao, J., 2019. A Global, 0.05-Degree Product of Solar-Induced Chlorophyll Fluorescence Derived from OCO-2, MODIS, and Reanalysis Data. Remote Sensing 11, 517. https://doi.org/10.3390/rs11050517

Lin, Y., Vosselman, G., Cao, Y., Yang, M.Y., 2020. Active and incremental learning for semantic ALS point cloud segmentation. ISPRS Journal of Photogrammetry and Remote Sensing 169, 73–92. https://doi.org/10.1016/j.isprsjprs.2020.09.003

Loshchilov, I., Hutter, F., 2017. Decoupled Weight Decay Regularization. https://doi.org/10.48550/ARXIV.1711.05101

Malone, S.L., Tulbure, M.G., Pérez-Luque, A.J., Assal, T.J., Bremer, L.L., Drucker, D.P., Hillis, V., Varela, S., Goulden, M.L., 2016. Drought resistance across California ecosystems: evaluating changes in carbon dynamics using satellite imagery. Ecosphere 7, e01561. https://doi.org/10.1002/ecs2.1561

Marshall, E., Randhir, T.O., 2008. Spatial modeling of land cover change and watershed response using Markovian cellular automata and simulation. Water Resources Research 44, 2006WR005514. https://doi.org/10.1029/2006WR005514

McDowell, N., Pockman, W.T., Allen, C.D., Breshears, D.D., Cobb, N., Kolb, T., Plaut, J., Sperry, J., West, A., Williams, D.G., Yepez, E.A., 2008. Mechanisms of plant survival and mortality during drought: why do some plants survive while others succumb to drought? New Phytologist 178, 719–739. https://doi.org/10.1111/j.1469-8137.2008.02436.x

Medellín-Azuara, J., Escriva-Bou, A., Gaudin, A.C.M., Schwabe, K.A., Sumner, D.A., 2024. Cultivating climate resilience in California agriculture: Adaptations to an increasingly volatile water future. Proc. Natl. Acad. Sci. U.S.A. 121, e2310079121. https://doi.org/10.1073/pnas.2310079121

Medellín-Azuara, J., Escriva-Bou, A., Rodríguez-Flores, J.M., Cole, S.A., Abatzoglou, J., Viers, J.H., Santos, N., Summer, D.A., Medina, C., Arévalo, R., Naumes, S., Bernacchi, L., 2022. Economic Impacts of the 2020–22 Drought on California Agriculture.

Meroni, M., Fasbender, D., Rembold, F., Atzberger, C., Klisch, A., 2019. Near real-time vegetation anomaly detection with MODIS NDVI: Timeliness vs. accuracy and effect of

anomaly computation options. Remote Sensing of Environment 221, 508–521. https://doi.org/10.1016/j.rse.2018.11.041

Mishra, A.K., Singh, V.P., 2010. A review of drought concepts. Journal of Hydrology 391, 202–216. https://doi.org/10.1016/j.jhydrol.2010.07.012

Moreno-Martínez, Á., Izquierdo-Verdiguier, E., Maneta, M.P., Camps-Valls, G., Robinson, N., Muñoz-Marí, J., Sedano, F., Clinton, N., Running, S.W., 2020. Multispectral high resolution sensor fusion for smoothing and gap-filling in the cloud. Remote Sensing of Environment 247, 111901. https://doi.org/10.1016/j.rse.2020.111901

Nelson, J.A., Walther, S., Jung, M., Gans, F., Kraft, B., Weber, U., Hamdi, Z., Duveiller, G., Zhang, W., Nelson, J.A., Walther, S., Jung, M., Gans, F., Kraft, B., Weber, U., Hamdi, Z., Duveiller, G., Zhang, W., 2023. FLUXCOM-X-BASE. https://doi.org/10.18160/5NZG-JMJE

Novick, K.A., Miniat, C.F., Vose, J.M., 2016. Drought limitations to leaf-level gas exchange: results from a model linking stomatal optimization and cohesion–tension theory. Plant Cell & Environment 39, 583–596. https://doi.org/10.1111/pce.12657

Oreshkin, B.N., Carpov, D., Chapados, N., Bengio, Y., 2019. N-BEATS: Neural basis expansion analysis for interpretable time series forecasting. https://doi.org/10.48550/ARXIV.1905.10437

Park, S., Im, J., Han, D., Rhee, J., 2020. Short-Term Forecasting of Satellite-Based Drought Indices Using Their Temporal Patterns and Numerical Model Output. Remote Sensing 12, 3499. https://doi.org/10.3390/rs12213499

Park, S., Im, J., Jang, E., Rhee, J., 2016. Drought assessment and monitoring through blending of multi-sensor indices using machine learning approaches for different climate regions. Agricultural and Forest Meteorology 216, 157–169. https://doi.org/10.1016/j.agrformet.2015.10.011

Pascanu, R., Mikolov, T., Bengio, Y., 2012. On the difficulty of training Recurrent Neural Networks. https://doi.org/10.48550/ARXIV.1211.5063

Pauloo, R.A., Escriva-Bou, A., Dahlke, H., Fencl, A., Guillon, H., Fogg, G.E., 2020. Domestic well vulnerability to drought duration and unsustainable groundwater management in California's Central Valley. Environ. Res. Lett. 15, 044010. https://doi.org/10.1088/1748-9326/ab6f10

Piotrowski, A.P., Napiorkowski, J.J., Piotrowska, A.E., 2020. Impact of deep learning-based dropout on shallow neural networks applied to stream temperature modelling. Earth-Science Reviews 201, 103076. https://doi.org/10.1016/j.earscirev.2019.103076

Prodhan, F.A., Zhang, J., Hasan, S.S., Pangali Sharma, T.P., Mohana, H.P., 2022. A review of machine learning methods for drought hazard monitoring and forecasting: Current research

trends, challenges, and future research directions. Environmental Modelling & Software 149, 105327. https://doi.org/10.1016/j.envsoft.2022.105327

Restaino, C., Young, D.J.N., Estes, B., Gross, S., Wuenschel, A., Meyer, M., Safford, H., 2019. Forest structure and climate mediate drought-induced tree mortality in forests of the Sierra Nevada, USA. Ecological Applications 29, e01902. https://doi.org/10.1002/eap.1902

Running, S., Mu, Q., Zhao, M., 2021. MODIS/Terra Gross Primary Productivity 8-Day L4 Global 500m SIN Grid V061. https://doi.org/10.5067/MODIS/MOD17A2H.061

Salakpi, E.E., Hurley, P.D., Muthoka, J.M., Bowell, A., Oliver, S., Rowhani, P., 2022. A dynamic hierarchical Bayesian approach for forecasting vegetation condition. Nat. Hazards Earth Syst. Sci. 22, 2725–2749. https://doi.org/10.5194/nhess-22-2725-2022

Scheffer, M., Carpenter, S., Foley, J.A., Folke, C., Walker, B., 2001. Catastrophic shifts in ecosystems. Nature 413, 591–596. https://doi.org/10.1038/35098000

Seddon, A.W.R., Macias-Fauria, M., Long, P.R., Benz, D., Willis, K.J., 2016. Sensitivity of global terrestrial ecosystems to climate variability. Nature 531, 229–232. https://doi.org/10.1038/nature16986

Tapley, B.D., Bettadpur, S., Watkins, M., Reigber, C., 2004. The gravity recovery and climate experiment: Mission overview and early results. Geophysical Research Letters 31, 2004GL019920. https://doi.org/10.1029/2004GL019920

Tchonkouang, R.D., Onyeaka, H., Nkoutchou, H., 2024. Assessing the vulnerability of food supply chains to climate change-induced disruptions. Science of The Total Environment 920, 171047. https://doi.org/10.1016/j.scitotenv.2024.171047

Thorne, J.H., Choe, H., Boynton, R.M., Bjorkman, J., Albright, W., Nydick, K., Flint, A.L., Flint, L.E., Schwartz, M.W., 2017. The impact of climate change uncertainty on California's vegetation and adaptation management. Ecosphere 8, e02021. https://doi.org/10.1002/ecs2.2021

Turco, M., Marcos-Matamoros, R., Castro, X., Canyameras, E., Llasat, M.C., 2019. Seasonal prediction of climate-driven fire risk for decision-making and operational applications in a Mediterranean region. Science of The Total Environment 676, 577–583. https://doi.org/10.1016/j.scitotenv.2019.04.296

Vermote, E., 2021. MODIS/Terra Surface Reflectance 8-Day L3 Global 500m SIN Grid V061. https://doi.org/10.5067/MODIS/MOD09A1.061

Vreugdenhil, M., Greimeister-Pfeil, I., Preimesberger, W., Camici, S., Dorigo, W., Enenkel, M., Van Der Schalie, R., Steele-Dunne, S., Wagner, W., 2022. Microwave remote sensing for agricultural drought monitoring: Recent developments and challenges. Front. Water 4, 1045451. https://doi.org/10.3389/frwa.2022.1045451

Watkins, M.M., Wiese, D.N., Yuan, D., Boening, C., Landerer, F.W., 2015. Improved methods for observing Earth's time variable mass distribution with GRACE using spherical cap mascons. JGR Solid Earth 120, 2648–2671. https://doi.org/10.1002/2014JB011547

West, H., Quinn, N., Horswell, M., 2019. Remote sensing for drought monitoring & impact assessment: Progress, past challenges and future opportunities. Remote Sensing of Environment 232, 111291. https://doi.org/10.1016/j.rse.2019.111291

Wiese, D.N., Landerer, F.W., Watkins, M.M., 2016. Quantifying and reducing leakage errors in the JPL RL05M GRACE mascon solution. Water Resources Research 52, 7490–7502. https://doi.org/10.1002/2016WR019344

Xiao, M., Koppa, A., Mekonnen, Z., Pagán, B.R., Zhan, S., Cao, Q., Aierken, A., Lee, H., Lettenmaier, D.P., 2017. How much groundwater did California's Central Valley lose during the 2012–2016 drought? Geophysical Research Letters 44, 4872–4879. https://doi.org/10.1002/2017GL073333

Young, D.J.N., Stevens, J.T., Earles, J.M., Moore, J., Ellis, A., Jirka, A.L., Latimer, A.M., 2017. Long-term climate and competition explain forest mortality patterns under extreme drought. Ecology Letters 20, 78–86. https://doi.org/10.1111/ele.12711

Zhang, D., Zhou, G., 2016. Estimation of Soil Moisture from Optical and Thermal Remote Sensing: A Review. Sensors 16, 1308. https://doi.org/10.3390/s16081308

Zhao, M., Heinsch, F.A., Nemani, R.R., Running, S.W., 2005. Improvements of the MODIS terrestrial gross and net primary production global data set. Remote Sensing of Environment 95, 164–176. https://doi.org/10.1016/j.rse.2004.12.011

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: